

2024, Mar. 2024 Release LOC:Inpatient Rehabilitation Transition Plan

Overview

Risk factors for readmission

Expected Discharge Level of Care

HOME

HOME CARE

SKILLED NURSING FACILITY (SNF)

SUBACUTE CARE (SAC)

LONG-TERM ACUTE CARE (LTAC)

ACUTE REHABILITATION

Notes

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Overview

Informational Notes

This guideline is intended to be used for those patients who are at increased risk of readmission following discharge. It serves as an adjunct to the review process and can assist the reviewer in **planning** for a safe discharge by providing a framework for identifying discharge needs and outlining the interventions necessary to ensure continuity of quality patient care. The National Quality Forum Consensus Standards Maintenance Committee has identified hospital discharge as a critical area for improvement (National Quality Forum (NQF). Safe Practices for Better Healthcare - 2010 Update. 2010 [cited Jan 4 2017]). It is recommended that clinicians be proactive and document a complete discharge plan for all patients and begin the planning process **at the time of admission**.

The term "transition of care" encompasses the patient who moves from one care setting to another, for example, transitioning from a skilled nursing or inpatient hospital setting to another care setting, such as the home. Transitioning care involves a collaborative effort between multiple providers, e.g., medical practitioner, nurse, social worker, pharmacist, and the patient, family and/or caregiver. To ensure quality of care, a transition should be appropriately timed and well coordinated (National Transitions of Care Coalition 2010; Jack et al., Ann Intern Med 2009; 150(3): 178-187).

Early comprehensive discharge planning and attention to transitioning care can reduce readmission rates, improve health outcomes, and may decrease length of stay (Jackson, Health Affairs 2013, 32: 1407-15; Osei-Anto et al., Health care leader action guide to reduce avoidable readmissions. 2010). Studies focusing on the transition of care from the inpatient to the outpatient setting have identified both safety and quality deficiencies in the current delivery system (Snow et al., J Gen Intern Med 2009, 24: 971-6; Coleman et al., Arch Intern Med 2006; 166(17): 1822-1828). Incomplete handoffs at the time of discharge can lead to adverse events for patients and result in avoidable readmissions (Halasyamani et al., J Hosp Med 2006; 1(6): 354-360). In 2013, the Center for Medicare and Medicaid Services updated its guidance for hospital discharge planning teams. The revised guidance establishes the medical information that must follow the patient at referral or transfer (Centers for Medicare & Medicaid Services, Hospital Compare datasets. 2016). As many as 20%(0.20) of the older patients are readmitted to the hospital within 30 days of discharge and the risk of readmission is increased when there is a delay in follow-up care (Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). In 2012, the Patient Protection and Affordable Care Act began financially penalizing hospitals for high readmission rates, making transition planning an even more vital component of discharge preparation. From July 2012 to June 2015, the Centers for Medicare and Medicaid Services' national overall readmission rate was 15.6 for all 30-day readmissions after discharges (Centers for Medicare & Medicaid Services, Hospital Compare datasets. 2016).

Readmission is a frequent and costly occurrence that is influenced by many factors including demographic, social, disease-related characteristics, medication related issues and perceived readiness for discharge. A recent study suggests that hospital readmissions within 30 days of discharge occur in 1 of 6 medical and 1 of 8 surgical Medicare patients (The Revolving Door: A Report on U.S. Hospital Readmissions. 2013). Discharge and transitional planning, which starts from the time of admission, is now starting even earlier in instances where hospital and community-based multidisciplinary teams work together alongside patients. Patient-centered care envisions patients as equal partners in this process (National Institute for Health and Care Excellence (NICE). Transition between inpatient hospital settings and community or care home settings for adults with social care needs. 2015).

Risk factors for readmission

● Adult

- Depression/Mood disorders ⁽¹⁾

● Schizophrenia ⁽²⁾

- Not receiving long acting antipsychotic injections ⁽³⁾

- Low health literacy ⁽⁴⁾

- Racial disparities ⁽⁵⁾

● Acute myocardial infarction (AMI)

- History of chronic obstructive airway disease (COPD) ⁽⁶⁾

- History of diabetes mellitus (DM) ⁽⁷⁾

- History of heart failure (HF) ⁽⁸⁾

- Need for hospice care ⁽⁹⁾

- Decreased renal function ⁽¹⁰⁾

- Low systolic blood pressure ⁽¹¹⁾

- Liver disease ⁽¹²⁾

- Infection ⁽¹³⁾

- Length of stay $\geq$ 8 days ⁽¹²⁾

- Discharge to home ⁽¹²⁾

- Coronary artery bypass graft (CABG) surgery and diabetes ⁽¹⁴⁾

● Heart failure (HF) ⁽¹⁵⁾

- Moderate to severe and age 65 or older ^(16, 17)

- Age 65 or younger at the time of initial HF admission ⁽¹⁸⁾

- High pre-discharge BNP level and less than 50% decrease from admission level ⁽¹⁹⁾

- Age 65 or older and post major non-cardiac surgery ⁽²⁰⁾

- Older individual and worsening renal function ⁽²¹⁾

- Length of stay $\geq$ 3 days ⁽²²⁾

- Psychosis ⁽²²⁾

- Comorbidities ⁽²³⁾

- No access to outpatient cardiac rehabilitation ⁽²⁴⁾

- Diabetes ⁽²⁵⁾

- Chronic obstructive pulmonary disease (COPD) ^(26, 27, 28, 29, 30, 31, 32, 33, 34)

● Pneumonia ⁽³⁵⁾

- Age 65 or older

- Worsening of comorbidities

- Instability at discharge ⁽³⁶⁾

- Treatment failure ⁽³⁶⁾

- Chronic liver disease ⁽³⁷⁾

- Sickle cell disease ⁽³⁸⁾

● Stroke/TIA ⁽³⁹⁾

- Hospital acquired infection ⁽⁴⁰⁾

- Diabetes mellitus ⁽⁴¹⁾

- Age 65 or older ⁽⁴²⁾

- 5 or more chronic comorbid conditions ⁽⁴³⁾

- Prior admission within 30 days ⁽⁴³⁾

- Age 60 or older, with comorbidity and history of adverse drug reaction ⁽⁴⁴⁾

● Discharge medications ^(45, 46)

- 6 or more

- High risk

● Age $\geq$ 75 ⁽⁴⁷⁾

- Comorbidity

- Functional disability

- Previous admission

- Prolonged length of stay

- Surgical patient ⁽⁴⁸⁾
 - Complex GI procedures
 - Total hip or knee arthroplasty ⁽⁴⁹⁾
 - Multiple comorbidities requiring pre-op admission
 - Surgical complications
- CMS Readmissions Reduction Program ⁽⁵⁰⁾
 - AMI
 - CABG
 - COPD, acute exacerbation
 - HF
 - Pneumonia
 - Including aspiration pneumonia
 - Sepsis patients coded with pneumonia present on admission (excluding severe sepsis)
 - Total hip or total knee arthroplasty ⁽⁴⁹⁾
- Pediatric
 - Asthma and prior asthma admission ⁽⁵¹⁾
 - Sickle Cell Disease ⁽⁵²⁾
 - Discharge less than 24 hours after transition from oxygen to room air
 - History of asthma
 - Greater than 3 admissions in the previous 12 months
 - Corticosteroid administration
 - Discharge from NICU
 - Bronchopulmonary dysplasia (BPD) ⁽⁵³⁾
 - Extreme low birth weight infants ⁽⁵⁴⁾
 - Sepsis ⁽⁵⁵⁾
 - Age less than 1 year
 - Neurologic or hematologic organ dysfunction during admission
 - Cardiovascular or bloodstream infection
 - Comorbidities
 - Congenital Heart Disease and length of stay greater than 14 days ⁽⁵⁶⁾
 - Medically complex child ^(57, 58)
 - ECMO ⁽⁵⁹⁾

Expected Discharge Level of Care

- HOME
 - Identify if home setting is appropriate
 - Assess for impairments that may impact self-care
 - Functional
 - Cognitive
 - Visual
 - Hearing
 - Speech
 - Language
 - Assess level of independence or if there is a willing and able caregiver
 - Complete prior to discharge
 - Condition specific discharge information
 - Acute Coronary Syndrome (ACS)
 - Follow-up appointments
 - Primary medical practitioner
 - Cardiologist
 - Referrals and consults
 - Community-based or inpatient cardiac rehabilitation program
 - Nutritional counseling for diet modification (e.g., low sodium, low fat)

- Palliative care or end of life support for advanced heart disease
- Smoking cessation counseling
- Patient Education
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations
 - Medication adherence ⁽⁶⁰⁾
 - Physical activity ⁽⁶¹⁾
 - Proper wound care, if post surgery (e.g., keeping wound clean, dressing wound and washing of hands when changing dressings) and notifying medical practitioner for signs of infections (e.g., wound area is more red or painful or warm to touch, blood, pus, or other fluid coming from the wound area, presence of fever, chills or muscle aches)
 - Recognizing acute cardiac symptoms and appropriate actions ⁽⁶²⁾
 - Recognizing signs of depression and seeking emotional support ⁽⁶³⁾
 - Secondary prevention methods ^(64, 65)
- Discharge medications
 - Ace inhibitor
 - Anticoagulant
 - Antihypertensive
 - Beta blocker
 - Statin
- Acute kidney failure
 - Follow-up appointments
 - Primary medical practitioner
 - Nephrologist
 - Referrals and consults
 - Nutrition consult
 - Smoking cessation counseling
 - Patient Education
 - Avoidance of nephrotoxins and adherence to treatment for conditions that contributed to kidney disease (e.g., hypertension, heart disease, high blood sugar)
 - Elimination or reduction of alcohol intake, if indicated
 - Following specific diet recommendations (including fluid intake) as instructed by medical practitioner
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations
 - Limiting over the counter medications containing sodium bicarbonate or sodium carbonate (e.g., laxatives, stomach medicine) and avoiding overuse of nonsteroidal anti-inflammatory drugs
 - Medication adherence
 - Monitoring sodium intake, if indicated
 - Strict medication adherence
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - Recognizing risk factors for developing chronic kidney failure and appropriate actions for follow up care (e.g., kidney function tests)
 - When to call medical practitioner when symptoms return or when blood sugar or blood pressure level is not within range recommended by medical practitioner
 - When to seek emergency care (e.g., sudden chest pain, trouble breathing)
 - Discharge medications
 - Aldosterone Antagonists
 - Anti-hypertensive
 - Calcimimetics
 - Diuretics
 - Erythropoiesis-stimulating agent
 - Phosphate binders
- Alcohol Dependence
 - Follow-up appointments

- Primary medical practitioner
- Behavioral health practitioners
- Referrals and consults
 - Consider palliative care or end of life support for advanced condition
 - Community support programs (e.g., Alcoholics Anonymous, Narcotics Anonymous)
 - Individual, group, or family counseling
 - Medication-assisted treatment
 - Sober living facility or transitional housing
 - Smoking cessation counseling
- Patient Education
 - Activity restrictions (e.g., not driving or operating machinery) as recommended by medical practitioner or pharmacist, especially when taking sedatives
 - Alcohol use disorder, dependence, and withdrawal symptoms
 - Avoidance of tobacco, excessive alcohol, and illicit drug use
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations
 - Community based resources and treatment options (e.g., medication management, cognitive behavioral therapy, motivational enhancement therapy, peer recovery coaching, twelve step meetings like Alcoholics Anonymous (AA) or Narcotics Anonymous (NA), non-twelve step support group meetings, or peer-to-peer guidance, sober living facility, transitional housing, residential treatment, SAMHSAs National Helpline))
 - Medication adherence
 - Alcohol use disorder, dependence, and withdrawal symptoms
 - Recognizing signs of depression or anxiety and seeking emotional support
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - When to call medical practitioner for new or worsening symptoms (e.g., severe shakiness, hallucinations, confusion, extreme drowsiness, increasing upper abdominal pain, uncontrolled vomiting)
 - When to seek emergency care (e.g., convulsion, difficulty breathing, chest pain, fast heart beat, heavy bleeding or vomiting blood, very drowsy or trouble awakening, fainting or loss of consciousness, thinking about hurting yourself or someone)
- Discharge medications
 - Acamprosate
 - Antiadrenergic agents
 - Anticonvulsants
 - Antidepressants
 - Antipsychotics
 - Benzodiazepines (Diazepam, Lorazepam)
 - Disulfiram
 - Opioid antagonists (Naltrexone)
- Arrhythmia, atrial
 - Follow-up appointments
 - Primary medical practitioner
 - Cardiologist
 - Referrals and consults
 - Smoking cessation counseling
 - Patient Education
 - Avoidance of tobacco, excessive alcohol, and illicit drug use
 - Managing ICD or pacemaker, if indicated (e.g., following precautions such as using caution with security devices, avoiding magnetic resonance image (MRI) scans, taking antibiotics before dental or surgical procedures, avoiding certain activities)
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations
 - Limiting intake of coffee, tea, cola, and over the counter meds with caffeine
 - Managing blood thinner medication, if indicated (blood tests to monitor drug level, preventing cuts, notifying medical practitioner for bleeding or bruising)

- Strict medication adherence
- Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
- Taking pulse and keeping record
- When to call medical practitioner for new or worsening symptoms (e.g., heart is beating too fast or slow, skipping beats or heart rate is more than 110 beats per minute, numbness in arm or hand on the same side if ICD/pacemaker inserted, signs of infection at surgical wound if indicated)
- When to seek emergency care (e.g., chest pain or discomfort in back, neck, jaw, stomach, or arm, shortness of breath, lightheadedness or a sudden cold sweat, feeling very tired, faint, or sick to your stomach, nausea, vomiting)
- Discharge medications
 - Antiarrhythmic
 - Anticoagulant
 - Antiplatelet
 - Digoxin
- Cellulitis
 - Follow-up appointments
 - Primary medical practitioner
 - Wound care Specialist
 - Referrals and consults
 - Home care for ongoing skilled needs
 - Smoking cessation counseling
 - Patient Education
 - Activity restrictions (e.g., not driving or operating machinery) as recommended by medical practitioner or pharmacist, especially when taking pain medicines or muscle relaxants
 - Care of the affected extremity (e.g., keeping the infected area clean, raising the infected area above the level of the heart to help decrease swelling)
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations
 - Recognizing factors or conditions that increase risk for infection and worsening symptoms (e.g., prolonged immobility, malnutrition, etc.)
 - Smoking cessation, if indicated
 - Strict medication adherence
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - When to call medical practitioner for new or worsening symptoms (vomiting, shaking chills, fever, difficulty or pain when moving joints above or below the infected area, increased redness, swelling, or drainage of pus from the infected area)
 - Discharge medications
 - Analgesic
 - Anti-infective
- COPD
 - Follow-up appointments
 - Primary medical practitioner
 - Pulmonologist
 - Referrals and consults
 - Enroll patient in call back program, if available ⁽⁶⁶⁾
 - Palliative care or end of life support for advanced COPD
 - Pulmonary rehab program
 - Refer patient to a medical home, disease management program, or case management program, if appropriate and available ^(67, 68)
 - Nutritional counseling or supplementation, if severely overweight or malnourished
 - Smoking cessation counseling
 - Patient Education
 - Avoidance of air pollutants, including a smoking cessation program, if indicated ⁽⁶⁹⁾

- Compliance with follow-up COPD care ^(70, 71)
- Correct use of inhalers ⁽⁷²⁾
- Energy conservation and breathing exercises
- Following instructions for oxygen therapy, if appropriate for chronic respiratory failure
- Influenza, pneumococcal, pertussis, and COVID-19 vaccinations ⁽⁷³⁾
- Medication adherence ⁽⁷⁴⁾
- Pulmonary rehabilitation ⁽⁷⁵⁾
- Recognizing signs of depression or anxiety and seeking emotional support from medical practitioner and community resources (e.g., Better Breathing Support Groups)
- Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
- When to call medical practitioner for new or worsening symptoms (e.g., chest pain when taking breaths, fever higher than 101.5 F (38.6 C), coughing up thick or blood-stained mucus, increased shortness of breath)
- When to seek emergency care (e.g., trouble talking or walking because of shortness of breath, bluish or gray colored lips or fingernails, trouble breathing that does not get easier with medicine, fast breathing or trouble catching breath, feeling like you are going to die, chest discomfort (pressure, fullness, squeezing or pain) that lasts more than a few minutes or goes away and comes back or chest discomfort that goes to your arms, neck, jaw or back)
- Discharge medications
 - Anticholinergics
 - Beta-Agonists
 - Inhaled glucocorticosteroids
 - Mucolytic Agents
 - Oral glucocorticosteroids (prednisone methylprednisone)
 - Phosphodiesterase -4 Inhibitors
- COVID-19
 - Follow-up appointments
 - Primary medical practitioner
 - Infectious disease specialist
 - Pulmonologist
 - Referrals and consults
 - Consider palliative care or end of life support for advanced illness
 - Medical home or home care for ongoing skilled needs
 - Nutritional counseling or supplementation, if severely overweight or malnourished
 - Smoking cessation counseling
 - Patient Education
 - Breathing exercises (e.g., incentive spirometry, rhythmic inhalation, coughing)
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations ⁽⁷⁶⁾
 - Limiting alcohol intake and drinking liquids as directed
 - Following instructions for oxygen therapy if appropriate for chronic respiratory failure
 - Following medical practitioner recommendations for transmission-based precautions (self-isolating in home and limiting contact with others to avoid spreading the virus) and when self-isolation can be eliminated depending upon symptoms, immune status and repeated testing
 - Knowledge of lingering COVID-19 symptoms (loss of taste and smell, nausea, or tiredness)
 - Medication adherence
 - Prevention measures, if immunocompromised (e.g., wearing a mask, staying 6ft apart from others outside not in living area, avoidance of crowds and poorly ventilated indoor spaces)
 - Recognizing signs of depression or anxiety and seeking emotional support
 - Smoking cessation program, if indicated ⁽⁷⁷⁾
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - When to seek emergency care (e.g., difficulty in breathing, chest pressure or pain, confusion, seizures, weakness or numbness of face or limbs, bluish lips or face, swelling of face or arms, fever)

- Discharge medications
 - Analgesic
 - Anti-infective or anti-viral or anti-fungal
- Gastrointestinal bleeding
 - Follow-up appointments
 - Primary medical practitioner
 - Gastroenterologist
 - Referrals and consults
 - Home care for ongoing skilled needs
 - Smoking cessation counseling
 - Patient Education
 - Anticoagulation management, if indicated
 - Avoidance of foods that may cause gastrointestinal irritation
 - Avoidance or caution when using nonsteroidal anti-inflammatory medications or aspirin
 - Elimination or reduction of alcohol intake if indicated
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations
 - Medication adherence
 - Smoking cessation program, if indicated ⁽⁷⁷⁾
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - When to seek emergency medical care (e.g., bright red blood in vomit that looks like coffee grounds, bowel movement with bright red blood, fainting)
 - Discharge medications
 - Analgesic
 - Anticoagulant
- Heart failure
 - Follow-up appointments
 - Primary medical practitioner
 - Cardiologist
 - Referrals and consults
 - Cardiac rehab program
 - Enroll patient in call back program, if available ⁽⁶⁶⁾
 - Nutritional counseling for diet modification (e.g., low sodium, low fat)
 - Palliative care or end of life support for advanced heart failure
 - Refer patient to a medical home, disease management program, or case management program, if appropriate and available ^(67, 68)
 - Smoking cessation counseling
 - Patient Education
 - Avoidance of tobacco, excessive alcohol, illicit drug, and nonsteroidal anti-inflammatory drug (NSAID) use
 - Energy conservation and breathing exercises
 - Fluid restriction, if ordered
 - Following instructions for oxygen therapy if appropriate for chronic respiratory failure
 - Following specific diet recommendations and maintaining a specific body weight
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations
 - Maintaining blood pressure within target range
 - Limiting salt (sodium) intake
 - Following medical practitioner recommendations for exercise and recognizing activities such as walking or gardening may be helpful
 - Knowledge of self-management and written action plan that identifies HF triggers and management of worsening symptoms (e.g., diuretic regimen, fluid restriction for volume overload)
 - Performing daily weights and reporting a gain of > 2 lbs/day to medical practitioner
 - Recognizing signs of depression or anxiety and seeking emotional support from medical practitioner and community resources (e.g., Together in HF from the American Association of HF)

Nurses, American Heart Association, and the American Heart Failure Society of America

- Strict medication adherence
- Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
- When to call medical practitioner for new or worsening symptoms (sudden weight gain of more than 2 pounds in 1 day or 5 pounds in 1 week, or weight gain parameters from medical practitioner, trouble breathing at rest, new or increased swelling of your legs or ankles, swelling or pain in your abdomen, breathing trouble at night, frequent coughing that doesn't go away)
- When to seek emergency care (e.g. severe shortness of breath, severe chest pain that does not resolve with rest or nitroglycerine, pink, foamy mucus with cough and shortness of breath, rapid or irregular heartbeat, passing out or fainting, etc.)
- Discharge medications
 - Aldosterone Antagonists
 - Angiotensin-Converting Enzyme (ACE)
 - Angiotensin II Receptor Blockers (or Inhibitors)
 - Angiotensin-Receptor Neprilysin Inhibitors (ARNIs)
 - Beta Blockers
 - Diuretics
 - Hydralazine and isosorbide dinitrate
- Osteomyelitis
 - Follow-up appointments
 - Primary medical practitioner
 - Orthopedic surgeon
 - Referrals and consults
 - Home care for ongoing skilled needs
 - Smoking cessation counseling
 - Patient Education
 - Activity restrictions (e.g., not driving or operating machinery) as recommended by medical practitioner or pharmacist, especially when taking pain medicines or muscle relaxants
 - Care of the affected extremity (e.g., keeping the area free from injury, use of a splint, sling, or brace as directed by medical practitioner)
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations
 - Proper care of catheter site, if indicated (e.g., keeping site clean and dry, monitoring for infection)
 - Proper wound care, if wound present (e.g., keeping wound clean, dressing wound and washing of hands when changing dressings, etc.)
 - Recognizing factors or conditions that increase risk for infection and worsening symptoms (e.g., prolonged immobility, malnutrition, etc.)
 - Smoking cessation, if indicated
 - Strict medication adherence
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - When to call medical practitioner for new or worsening symptoms (increasing pain, swelling, warmth, redness or drainage from the infected area, more pain around the infected area, fever)
 - Discharge medications
 - Analgesic
 - Antifungal
 - Anti-infective
- Pneumonia
 - Follow-up appointments
 - Primary medical practitioner
 - Pulmonologist
 - Referrals and consults
 - Home care for patients with ongoing skilled needs
 - Consider palliative care or end of life support for advanced conditions contributing to pneumonia

- Smoking cessation counseling
- Patient Education
 - Breathing exercises (e.g., incentive spirometry, rhythmic inhalation and coughing)
 - Drinking liquids to stay hydrated to promote coughing up mucus (unless told to limit fluids)
 - Influenza, pneumococcal, pertussis and COVID-19 vaccinations ⁽⁷⁶⁾
 - Recognizing factors or conditions that increase risk for pneumonia or worsen symptoms (e.g., smoking and air pollution, upper respiratory infections, prolonged immobility, malnutrition, poor handwashing, etc) and
 - Following instructions for oxygen therapy, if appropriate for chronic respiratory failure
 - Medication adherence
 - Smoking cessation, if indicated ⁽⁷⁸⁾
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - Use of a cool mist humidifier to increase air moisture
 - When to call medical practitioner for new or worsening symptoms (e.g., shortness of breath, chest pain when you take a breath, wheezing, coughing up mucus that is thick or blood-stained, fever higher than 101.5 F (38.6 C))
- Discharge medications
 - Analgesic
 - Anti-infective, anti-viral, or anti-fungal
 - Inhaled glucocorticosteroids
 - Mucolytic Agents
 - Oral glucocorticosteroids
- Pyelonephritis or complex UTI
 - Follow-up appointments
 - Primary medical practitioner
 - Urologist
 - Referrals and consults
 - Home care for ongoing skilled needs
 - Smoking cessation counseling
 - Patient Education
 - Influenza, pneumococcal, pertussis and COVID-19 vaccinations
 - Knowledge of factors or conditions that increase risk for infection and worsen symptoms (e.g., prolonged immobility, malnutrition, etc.)
 - Limiting alcohol intake and drinking enough fluids to keep urine light yellow in color (unless told to limit fluids)
 - Medication adherence and completing full antibiotic course
 - Preventing future infection (keeping genital area clean, proper wiping of genital, urinating frequently, practicing safe sex)
 - Proper care of indwelling catheters, if applicable (including clean procedures and when to replace)
 - Smoking cessation, if indicated
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - When to call health care medical practitioner for new or worsening symptoms (fever, shaking chills, vomiting, urinating very little or not at all, blood in your urine, dark-colored or foul-smelling urine, side or back pain that gets worse)
 - Discharge medications
 - Analgesic
 - Anti-infective
- Respiratory failure
 - Follow-up appointments
 - Primary medical practitioner
 - Pulmonologist
 - Referrals and consults

- Enroll patient in call back program, if available ⁽⁶⁶⁾
- Palliative care or end of life support for advanced lung disease
- Refer patient to a medical home, disease management program, or case management program, if appropriate and available ^(67, 68)
- Nutritional counseling or supplementation, if severely overweight or malnourished
- Smoking cessation counseling
- Patient Education
 - Avoidance of air pollutants, including a smoking cessation program, if indicated ⁽⁶⁹⁾
 - Breathing exercises to manage shortness of breath (e.g., pursed-lip breathing)
 - Following instructions for oxygen therapy, if appropriate for chronic respiratory failure
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations ⁽⁷³⁾
 - Recognizing signs of depression or anxiety and seeking emotional support from medical practitioner and community resources (e.g., Better Breathing Support Groups)
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - When to call medical practitioner for new or worsening symptoms (e.g., increased swelling in legs, feet or abdomen, wheezing, coughing up blood in sputum, fever)
 - When to seek emergency care (e.g., difficulty breathing, shortness of breath, chest pain)
- Discharge medications
 - Anticholinergics
 - Beta-Agonists
 - Inhaled glucocorticosteroids
 - Mucolytic Agents
 - Oral glucocorticosteroids (prednisone methylprednisone)
 - Phosphodiesterase -4 Inhibitors
- Sepsis
 - Follow-up appointments
 - Primary medical practitioner
 - Infectious disease specialist
 - Wound care specialist
 - Referrals and consults
 - Home care for ongoing skilled needs
 - Consider palliative care or end of life support for advanced conditions contributing to sepsis
 - Smoking cessation counseling
 - Patient Education
 - Influenza, pneumococcal, pertussis and COVID-19 vaccinations
 - Knowledge of factors or conditions that increase risk for infection and worsen symptoms (e.g., prolonged immobility, malnutrition)
 - Limiting alcohol intake and drinking enough fluids to keep urine light yellow in color, unless told to limit fluids
 - Medication adherence
 - Proper wound care, if wound present or post surgery (e.g., keeping wound clean, dressing wound and washing of hands when changing dressings, etc.)
 - Smoking cessation, if indicated
 - Staying physically active with exercise as instructed by medical practitioner
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - Understanding Post Sepsis Syndrome and long-term effects of sepsis (e.g., insomnia, extreme fatigue, poor concentration, decreased mental or cognitive functioning, loss of self-esteem, nightmares, vivid hallucinations, panic attacks, disabling muscle and joint pain)
 - When to call medical practitioner for new or worsening symptoms (chills, sweats, nausea or vomiting, fever higher than 101.5 F (38.6 C), body aches, fast heartbeat, dizziness when you stand up or fainting, confusion, redness, swelling, or drainage of pus from any wounds or from around area where an IV goes into skin)

- Discharge medications
 - Analgesic
 - Anti-infective, anti-viral, or anti-fungal
- Stroke
 - Follow-up appointments
 - Primary medical practitioner
 - Neurologist
 - Referrals and consults
 - Home or community-based rehabilitation or inpatient rehabilitation program
 - Palliative care or end of life support for advanced disease
 - Smoking cessation counseling
 - Patient Education
 - Achieving and maintaining a healthy weight
 - Elimination or reduction of alcohol intake
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations
 - Recognizing depression is common after stroke and seeking emotional support
 - Smoking cessation and avoiding passive smoke ⁽⁷⁹⁾
 - Staying physically active as advised by medical practitioner
 - Strict medication adherence ⁽⁸⁰⁾
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - Understanding risk of having another stroke and any necessary life-style modification(s) (e.g., maintaining a healthy weight and blood pressure, quitting smoking, and for ischemic stroke healthy eating to control blood sugar and cholesterol level)
 - Understanding risk of falling and changes necessary to maintain a safe living environment if indicated
 - Understanding common problems associated with stroke (e.g., changes in behavior, difficulty doing tasks, memory, moving and doing ADLs, muscle spasms, swallowing, lack of sensation or awareness of one part of the body, thinking, talking, etc.) and any planned therapy services (e.g., physical, occupational or speech therapy)
 - When to call medical practitioner for new or worsening symptoms (e.g., loss of balance or coordination, personality changes, dizziness or lightheadedness, headache that will not go away after treatment, confusion, forgetfulness)
 - When to seek emergency care (e.g., weakness, numbness, drooping, or tingling of face, trouble seeing, severe headache, seizure, trouble thinking, trouble swallowing, trouble talking or understanding speech, feeling dizzy, weakness, numbness or tingling in arm or leg, trouble walking or moving arm or leg, loss of bladder or bowel control)
 - Discharge medications
 - Anticoagulant
 - Antihypertensive
 - Antiplatelet agent
- Assess social determinants of health (SDOH)
 - Adult ⁽⁸¹⁾
 - Assess for alcohol or substance use
 - Care coordination with outpatient medical or behavioral health providers
 - Provide information about community resources and programs
 - Referral to addiction social support systems
 - Assess for domestic abuse, abuse of elderly, and vulnerable adults ⁽⁸²⁾
 - Provide information about legal and community resources
 - Create a long-term safety plan or make appropriate referral
 - Consider intensive home nursing interventions and psychosocial counseling interventions ⁽⁸³⁾
 - Assess need for end of life planning
 - Consider palliative care for patients with end stage disease
 - Consider hospice care for terminally ill patients with a life expectancy of less than 6 months ⁽⁸⁴⁾

- Assess for strained or limited financial resources
 - For income insecurity, identify income-support resources ⁽⁸⁵⁾
 - Consider direct income assistance benefits and indirect income assistance programs (i.e., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)
 - Consider referral to Area Agencies on Aging (AAA) for information on nutrition and meals programs, home homemaking services, transportation resources
 - For food insecurity provide information about local food banks and Supplemental Nutrition Assistance Program (SNAP) incentive programs ⁽⁸⁶⁾
- Assess for functional or ADL impairment
 - Consider an alternate level of care
 - Assess the need for a capacity evaluation ⁽⁸⁷⁾
 - Provide information to patient and family about advance directive or power of attorney
- Assess for mental health co-morbidities
 - Arrange for appointment with primary care provider or behavioral health specialist upon discharge
 - Coordinate care with outpatient behavioral health providers
 - Provide information on community resources
- Assess risk for readmission
 - Consider home care services
 - Consider transfer to an alternative level of care or readmission reduction program ⁽⁸⁸⁾
 - Consider telehealth and mobile health interventions
 - Provide patient and family with information on disease management programs
 - For older individuals at risk for malnutrition consider dietitian evaluation, home-delivered meals, and other community resources ⁽⁸⁹⁾
- Assess risk for treatment nonadherence
 - For patients with poor insight into illness or noncompliance with treatment, introduce peer support groups to help improve service user experience and quality of life (National Institute for Health and Care Excellence (NICE) ⁽⁹⁰⁾
 - Consider in-home intervention to prevent medical relapse, or readmission, and to enhance medication compliance
 - For patients at risk of medication nonadherence, consider patient education and behavioral support initiatives such as text message reminders to take a medication and electronic medication dispensers
- Assess for environmental exposure to toxic substances and other physical hazards
 - For tobacco use and/or exposure, information should be given about tobacco-use cessation and the risk of second-hand smoke ⁽⁹¹⁾
 - For lead poisoning, provide community resources for primary prevention and secondary prevention strategies including blood lead level testing and follow up ⁽⁹²⁾
- Assess for unstable living environment, housing vulnerability or potential for homelessness upon discharge ⁽⁹³⁾
 - Assist or provide health insurance application
 - Assist or provide homeless shelter application and provide information to patient
 - Consider referral to walk-in state or federally sponsored outpatient programs
 - Consider referral to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system ⁽⁹⁴⁾
 - For individuals with homelessness and the greatest service need (i.e., serious mental illness, history of hospitalizations, functional impairment) consider Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community-Based Treatment services ⁽⁹⁵⁾
- Assess for unsafe home environment
 - If home environment is unsafe, consider a skilled nursing facility (SNF)
 - For patients at risk for falls, provide information on community-based fall prevention programs through the National Council on Aging
- Pediatric ⁽⁹⁶⁾

- Assess for alcohol or substance use
 - Coordinate care with outpatient primary care or behavioral health providers
 - Refer to addiction social support systems
 - Provide information about community resources and programs for family-centered treatment (e.g., intensive case management, detox, outpatient, residential treatment, day treatment, and medication assisted treatment)
 - Provide information about home visitation programs (e.g., Parent-Child Assistance Program (PCAP) for pregnant and parenting individual with substance use disorders aftercare) ⁽⁹⁷⁾
 - Provide information about Cognitive Behavioral Therapy (CBT) Multidimensional Family Therapy (MDFT) and/or Pharmacotherapy for youth with co-occurring emotional/mental disorders ⁽⁹⁸⁾
- Assess for child maltreatment or abuse (e.g., sexual and/or psychological abuse)
 - Arrange for alternative housing situation, foster care, or shelter if home environment is unsafe
 - Assist with creating a long-term safety plan or make appropriate referral
 - Refer to child protective services (CPS) and other domestic violence services
 - Refer for psychosocial support
 - Provide community resources and legal information
 - Provide information on parent training programs such as the Nurse Family Partnership Program (NFP) ⁽⁹⁹⁾
- Assess need for end of life planning, palliative care, or hospice services
 - Consider palliative care for children with end stage disease
 - Consider hospice care for terminally ill children with a life expectancy of less than 6 months
- Assess for environmental exposure to toxic substances and other physical hazards
 - For lead poisoning provide community resources for primary prevention and secondary prevention strategies (e.g., blood lead level testing and follow up) ⁽⁹²⁾
 - For tobacco use (e.g., cigarettes, E-cigarettes, vaping) provide education and/or brief counseling about tobacco use cessation and the risk of second-hand smoke ⁽⁹¹⁾
- Assess for infants and children with special health care needs
 - Provide anticipatory guidance and education to caregivers regarding medical condition and expected progress after discharge over the next few days, months, or years, as appropriate ⁽¹⁰⁰⁾
 - Consider referral for follow-up care to neonatologist or other medical subspecialist for infants or children with ongoing and/or chronic medical issues, (e.g., bronchopulmonary dysplasia or feeding dysfunction) ⁽¹⁰¹⁾
 - Initiate referral for adaptive or assistive medical equipment and equipment for technology-dependent children (e.g., home oxygen, ventilators, monitors, or tube feeding) ⁽¹⁰¹⁾
 - Consider referral for neurodevelopmental assessment for high-risk infants
 - Consider referral to post-discharge follow-up programs ⁽¹⁰²⁾
 - Consider referral for behavioral support and/or counseling for caregivers through therapy groups, telephone support, and/or peer support groups for caregivers of children with similar health needs ⁽¹⁰³⁾
 - Consider referral to lactation consultant in the hospital and home visits for infants at risk for poor lactation or suboptimal infant breast-feeding behavior ⁽¹⁰⁴⁾
 - Consider referral for respite services
- Assess for strained or limited financial resources
 - For food insecurity provide information about local food banks and special nutrition programs (e.g., Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), U.S. Department of Agriculture Summer Food Service ⁽⁸⁶⁾
 - For income insecurity identify income resources and refer to direct and/or indirect income assistance programs (e.g., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)
- Assess for mental health co-morbidities
 - Arrange appointment with primary care provider or behavioral health specialist upon discharge
 - Coordinate care with outpatient behavioral health providers
 - Provide information about suicide prevention, including the National Suicide Prevention hotline ⁽¹⁰⁵⁾
 - Consider referral for psychiatric services for suicide risk screening and intervention ⁽¹⁰⁵⁾

- Consider referral to school-based mental health interventions ⁽¹⁰⁶⁾
- Provide information about Intensive Community-based treatment (e.g., Functional Family Therapy, Intensive Child and Adolescent and Psychiatric Services, Multidimensional Family Therapy, and Intensive Case Management)
- Consider referral to day treatment program, residential addiction services, or partial hospitalization
- Assess risk for treatment nonadherence
 - Provide information about educational and behavioral support initiatives for children at risk of medication nonadherence and their caregivers ⁽¹⁰⁷⁾
 - Provide information about peer support groups for children and/or families with poor insight into illness or noncompliance with treatment ⁽¹⁰⁸⁾
 - Consider home care services or community resources to prevent medical relapse, or readmission, and to enhance medication compliance
- Assess risk for readmission
 - Consider transfer to an alternative level of care such as subacute care
 - Consider home care services
 - Consider early childhood home visiting programs (e.g., Early Head Start; Early On; Nurse Family Partnership; and the Maternal, Infant, and Early Childhood Home Visiting Program) ⁽¹⁰²⁾
 - Consider telehealth and web-based training programs (e.g., telepsychiatry or telebehavioral health and mobile health interventions) ⁽¹⁰⁹⁾
 - Consider intensive in-home behavioral health service
 - Provide patient and family with condition-specific information including appropriate planning for prevention of RSV infection and appropriate immunizations if appropriate
- Assess for special health risk behaviors in adolescence and young adulthood
 - Arrange follow-up appointment with primary care provider and/or specialists upon discharge
 - Coordinate care with outpatient behavioral health providers
 - Provide information on mentoring programs, (e.g., Big Brothers Big Sisters of America)
 - Consider referral for Intensive Community-Based treatment for adolescents with comorbid behavioral health issues
 - Provide community resources for teen pregnancy prevention and family planning (use of birth control, subsequent pregnancies) ⁽¹¹⁰⁾
- For eating disorders
 - Provide information on nutrition services including education and resources for diet management, physical activity, and promotion of healthy eating attitudes and behaviors ⁽¹¹¹⁾
 - Provide information on outpatient psychosocial interventions and Family-Based Treatment ⁽¹¹²⁾
 - Refer to an intensive weight management program that includes culturally tailored interventions ⁽¹¹³⁾
- Provide parenting resources
 - Provide information on Family Resource Centers and Parenting support groups for young, first-time parents, (e.g., National parent help line) ⁽¹¹⁴⁾
 - Provide information on community resources to improve parenting skills (face-to-face or by telephone) ⁽¹¹⁵⁾
- Assess for unstable living environment, housing vulnerability or potential for homelessness upon discharge ⁽⁹³⁾
 - Assist or provide health insurance application
 - Assist or provide homeless shelter application to caregivers
 - Consider referral to local housing coordinator or case manager for immediate access to permanent supportive housing and coordinated access system ⁽⁹⁴⁾
 - Consider referral to walk-in state or federally sponsored outpatient programs
 - Consider referral to Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community Treatment services ⁽¹¹⁶⁾
- Plan for follow-up care ^(117, 118)
 - Assist or arrange follow-up care or testing
 - Solicit input from patient or caregiver about convenient dates, times, and places

- Coordinate appointments with practitioners, testing sites, and other service areas
- Discuss reason for, and importance of, follow-up care
- Organize post discharge services and medical supplies or equipment
 - Ensure patient or caregiver understands the importance of services
 - Discuss what to expect with each service
- Clarify any questions about medical care and follow-up
- Refer patient to a medical home, disease management program, or case management program, if appropriate and available ^(67, 68)
- Enroll patient in call back program, if available ⁽⁶⁶⁾
- Review discharge instructions ⁽¹¹⁹⁾
 - Identify any pending in-hospital tests or studies and determine who will follow-up with them ⁽¹²⁰⁾
 - Educate patient or caregiver about relevant diagnoses ⁽¹²¹⁾
 - Review contact information for medical practitioner, specialty care, services, and outpatient testing
 - Provide a written list of scheduled appointments and tests
 - Assess understanding of plan by asking patient to explain it in his or her own words (may require discussion with caregiver) ^(122, 123)
- Medication reconciliation ⁽⁴⁶⁾
 - Provide list and discuss
 - How and when medications should be taken
 - Reasons for medications
 - Common side effects
 - Changes to previously prescribed medications
 - Review of over-the-counter (OTC) medications, supplements, and herbal remedies
 - Assess for potential drug interactions or drug allergies
 - Assess the risks and benefits of opiate therapy for acute/chronic pain management, following recommendations for initiation, continuation, and follow-up management of opioid therapy ⁽¹²⁴⁾
 - Address potential reasons for non-adherence ⁽¹²⁵⁾
 - Does the patient have a mechanism for filling prescriptions?
 - Does the patient have the means to pay for medications?
 - Are there community resources or discount programs available?
 - Are there generic alternatives?
 - Ensure prescriptions are provided or e-prescriptions sent
 - Ensure patient or caregiver verbalizes understanding of regimen ⁽¹²²⁾
- Assess understanding of when and where to seek help ⁽¹²²⁾
 - What constitutes an emergency
 - What to do in the case of an emergency
 - When to contact medical practitioner
 - Who to call on weekends or evenings
- Identify and address transportation needs
- Obtain discharge summary and ensure it is transmitted to medical practitioner prior to the first follow-up visit ^(126, 127)
- HOME CARE
 - Identify if home setting is appropriate
 - Assess for impairments that may impact self-care
 - Functional
 - Cognitive
 - Visual
 - Hearing
 - Speech
 - Language
 - Assess level of independence or if there is a willing and able caregiver
 - Assess need for skilled home care services for assessment, treatment, monitoring, or education
 - Arrange home care services ⁽¹²⁸⁾

- Set expectation that home care services are temporary while skilled need exists
- Discuss reason for, and importance of, home health services

● Complete prior to discharge**● Condition specific discharge information****● Acute Coronary Syndrome (ACS)****● Follow-up appointments**

- Primary medical practitioner
- Cardiologist

● Referrals and consults

- Community-based or inpatient cardiac rehabilitation program
- Nutritional counseling for diet modification (e.g., low sodium, low fat)
- Palliative care or end of life support for advanced heart disease
- Smoking cessation counseling

● Patient Education

- Influenza, pneumococcal, pertussis, and COVID-19 vaccinations
- Medication adherence ⁽⁶⁰⁾
- Physical activity ⁽⁶¹⁾
- Proper wound care, if post surgery (e.g., keeping wound clean, dressing wound and washing of hands when changing dressings) and notifying medical practitioner for signs of infections (e.g., wound area is more red or painful or warm to touch, blood, pus, or other fluid coming from the wound area, presence of fever, chills or muscle aches)
- Recognizing acute cardiac symptoms and appropriate actions ⁽⁶²⁾
- Recognizing signs of depression and seeking emotional support ⁽⁶³⁾
- Secondary prevention methods ^(64, 65)

● Discharge medications

- Ace inhibitor
- Anticoagulant
- Antihypertensive
- Beta blocker
- Statin

● Acute kidney failure**● Follow-up appointments**

- Primary medical practitioner
- Nephrologist

● Referrals and consults

- Nutrition consult
- Smoking cessation counseling

● Patient Education

- Avoidance of nephrotoxins and adherence to treatment for conditions that contributed to kidney disease (e.g., hypertension, heart disease, high blood sugar)
- Elimination or reduction of alcohol intake, if indicated
- Following specific diet recommendations (including fluid intake) as instructed by medical practitioner
- Influenza, pneumococcal, pertussis, and COVID-19 vaccinations
- Limiting over the counter medications containing sodium bicarbonate or sodium carbonate (e.g., laxatives, stomach medicine) and avoiding overuse of nonsteroidal anti-inflammatory drugs
- Medication adherence
- Monitoring sodium intake, if indicated
- Strict medication adherence
- Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
- Recognizing risk factors for developing chronic kidney failure and appropriate actions for follow up care (e.g., kidney function tests)
- When to call medical practitioner when symptoms return or when blood sugar or blood pressure

- level is not within range recommended by medical practitioner
 - When to seek emergency care (e.g., sudden chest pain, trouble breathing)
- Discharge medications
 - Aldosterone Antagonists
 - Anti-hypertensive
 - Calcimimetics
 - Diuretics
 - Erythropoiesis-stimulating agent
 - Phosphate binders
- Alcohol Dependence
 - Follow-up appointments
 - Primary medical practitioner
 - Behavioral health practitioners
 - Referrals and consults
 - Consider palliative care or end of life support for advanced condition
 - Community support programs (e.g., Alcoholics Anonymous, Narcotics Anonymous)
 - Individual, group, or family counseling
 - Medication-assisted treatment
 - Sober living facility or transitional housing
 - Smoking cessation counseling
 - Patient Education
 - Activity restrictions (e.g., not driving or operating machinery) as recommended by medical practitioner or pharmacist, especially when taking sedatives
 - Alcohol use disorder, dependence, and withdrawal symptoms
 - Avoidance of tobacco, excessive alcohol, and illicit drug use
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations
 - Community based resources and treatment options (e.g., medication management, cognitive behavioral therapy, motivational enhancement therapy, peer recovery coaching, twelve step meetings like Alcoholics Anonymous (AA) or Narcotics Anonymous (NA), non-twelve step support group meetings, or peer-to-peer guidance, sober living facility, transitional housing, residential treatment, SAMHSAs National Helpline))
 - Medication adherence
 - Alcohol use disorder, dependence, and withdrawal symptoms
 - Recognizing signs of depression or anxiety and seeking emotional support
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - When to call medical practitioner for new or worsening symptoms (e.g., severe shakiness, hallucinations, confusion, extreme drowsiness, increasing upper abdominal pain, uncontrolled vomiting)
 - When to seek emergency care (e.g., convulsion, difficulty breathing, chest pain, fast heart beat, heavy bleeding or vomiting blood, very drowsy or trouble awakening, fainting or loss of consciousness, thinking about hurting yourself or someone)
 - Discharge medications
 - Acamprosate
 - Antiadrenergic agents
 - Anticonvulsants
 - Antidepressants
 - Antipsychotics
 - Benzodiazepines (Diazepam, Lorazepam)
 - Disulfiram
 - Opioid antagonists (Naltrexone)
 - Arrhythmia, atrial
 - Follow-up appointments
 - Primary medical practitioner

- Cardiologist
- Referrals and consults
 - Smoking cessation counseling
- Patient Education
 - Avoidance of tobacco, excessive alcohol, and illicit drug use
 - Managing ICD or pacemaker, if indicated (e.g., following precautions such as using caution with security devices, avoiding magnetic resonance image (MRI) scans, taking antibiotics before dental or surgical procedures, avoiding certain activities)
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations
 - Limiting intake of coffee, tea, cola, and over the counter meds with caffeine
 - Managing blood thinner medication, if indicated (blood tests to monitor drug level, preventing cuts, notifying medical practitioner for bleeding or bruising)
 - Strict medication adherence
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - Taking pulse and keeping record
 - When to call medical practitioner for new or worsening symptoms (e.g., heart is beating too fast or slow, skipping beats or heart rate is more than 110 beats per minute, numbness in arm or hand on the same side if ICD/pacemaker inserted, signs of infection at surgical wound if indicated)
 - When to seek emergency care (e.g., chest pain or discomfort in back, neck, jaw, stomach, or arm, shortness of breath, lightheadedness or a sudden cold sweat, feeling very tired, faint, or sick to your stomach, nausea, vomiting)
- Discharge medications
 - Antiarrhythmic
 - Anticoagulant
 - Antiplatelet
 - Digoxin
- Cellulitis
 - Follow-up appointments
 - Primary medical practitioner
 - Wound care Specialist
 - Referrals and consults
 - Home care for ongoing skilled needs
 - Smoking cessation counseling
 - Patient Education
 - Activity restrictions (e.g., not driving or operating machinery) as recommended by medical practitioner or pharmacist, especially when taking pain medicines or muscle relaxants
 - Care of the affected extremity (e.g., keeping the infected area clean, raising the infected area above the level of the heart to help decrease swelling)
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations
 - Recognizing factors or conditions that increase risk for infection and worsening symptoms (e.g., prolonged immobility, malnutrition, etc.)
 - Smoking cessation, if indicated
 - Strict medication adherence
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - When to call medical practitioner for new or worsening symptoms (vomiting, shaking chills, fever, difficulty or pain when moving joints above or below the infected area, increased redness, swelling, or drainage of pus from the infected area)
 - Discharge medications
 - Analgesic
 - Anti-infective
- COPD
 - Follow-up appointments

- Primary medical practitioner
- Pulmonologist
- Referrals and consults
 - Enroll patient in call back program, if available ⁽⁶⁶⁾
 - Palliative care or end of life support for advanced COPD
 - Pulmonary rehab program
 - Refer patient to a medical home, disease management program, or case management program, if appropriate and available ^(67, 68)
 - Nutritional counseling or supplementation, if severely overweight or malnourished
 - Smoking cessation counseling
- Patient Education
 - Avoidance of air pollutants, including a smoking cessation program, if indicated ⁽⁶⁹⁾
 - Compliance with follow-up COPD care ^(70, 71)
 - Correct use of inhalers ⁽⁷²⁾
 - Energy conservation and breathing exercises
 - Following instructions for oxygen therapy, if appropriate for chronic respiratory failure
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations ⁽⁷³⁾
 - Medication adherence ⁽⁷⁴⁾
 - Pulmonary rehabilitation ⁽⁷⁵⁾
 - Recognizing signs of depression or anxiety and seeking emotional support from medical practitioner and community resources (e.g., Better Breathing Support Groups)
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - When to call medical practitioner for new or worsening symptoms (e.g., chest pain when taking breaths, fever higher than 101.5 F (38.6 C), coughing up thick or blood-stained mucus, increased shortness of breath)
 - When to seek emergency care (e.g., trouble talking or walking because of shortness of breath, bluish or gray colored lips or fingernails, trouble breathing that does not get easier with medicine, fast breathing or trouble catching breath, feeling like you are going to die, chest discomfort (pressure, fullness, squeezing or pain) that lasts more than a few minutes or goes away and comes back or chest discomfort that goes to your arms, neck, jaw or back)
- Discharge medications
 - Anticholinergics
 - Beta-Agonists
 - Inhaled glucocorticosteroids
 - Mucolytic Agents
 - Oral glucocorticosteroids (prednisone methylprednisone)
 - Phosphodiesterase -4 Inhibitors
- COVID-19
 - Follow-up appointments
 - Primary medical practitioner
 - Infectious disease specialist
 - Pulmonologist
 - Referrals and consults
 - Consider palliative care or end of life support for advanced illness
 - Medical home or home care for ongoing skilled needs
 - Nutritional counseling or supplementation, if severely overweight or malnourished
 - Smoking cessation counseling
 - Patient Education
 - Breathing exercises (e.g., incentive spirometry, rhythmic inhalation, coughing)
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations ⁽⁷⁶⁾
 - Limiting alcohol intake and drinking liquids as directed
 - Following instructions for oxygen therapy if appropriate for chronic respiratory failure
 - Following medical practitioner recommendations for transmission-based precautions (self-

- isolating in home and limiting contact with others to avoid spreading the virus) and when self-isolation can be eliminated depending upon symptoms, immune status and repeated testing
- Knowledge of lingering COVID-19 symptoms (loss of taste and smell, nausea, or tiredness)
 - Medication adherence
 - Prevention measures, if immunocompromised (e.g., wearing a mask, staying 6ft apart from others outside not in living area, avoidance of crowds and poorly ventilated indoor spaces)
 - Recognizing signs of depression or anxiety and seeking emotional support
 - Smoking cessation program, if indicated ⁽⁷⁷⁾
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - When to seek emergency care (e.g., difficulty in breathing, chest pressure or pain, confusion, seizures, weakness or numbness of face or limbs, bluish lips or face, swelling of face or arms, fever)
- Discharge medications
 - Analgesic
 - Anti-infective or anti-viral or anti-fungal
 - Gastrointestinal bleeding
 - Follow-up appointments
 - Primary medical practitioner
 - Gastroenterologist
 - Referrals and consults
 - Home care for ongoing skilled needs
 - Smoking cessation counseling
 - Patient Education
 - Anticoagulation management, if indicated
 - Avoidance of foods that may cause gastrointestinal irritation
 - Avoidance or caution when using nonsteroidal anti-inflammatory medications or aspirin
 - Elimination or reduction of alcohol intake if indicated
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations
 - Medication adherence
 - Smoking cessation program, if indicated ⁽⁷⁷⁾
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - When to seek emergency medical care (e.g., bright red blood in vomit that looks like coffee grounds, bowel movement with bright red blood, fainting)
 - Discharge medications
 - Analgesic
 - Anticoagulant
 - Heart failure
 - Follow-up appointments
 - Primary medical practitioner
 - Cardiologist
 - Referrals and consults
 - Cardiac rehab program
 - Enroll patient in call back program, if available ⁽⁶⁶⁾
 - Nutritional counseling for diet modification (e.g., low sodium, low fat)
 - Palliative care or end of life support for advanced heart failure
 - Refer patient to a medical home, disease management program, or case management program, if appropriate and available ^(67, 68)
 - Smoking cessation counseling
 - Patient Education
 - Avoidance of tobacco, excessive alcohol, illicit drug, and nonsteroidal anti-inflammatory drug (NSAID) use
 - Energy conservation and breathing exercises

- Fluid restriction, if ordered
- Following instructions for oxygen therapy if appropriate for chronic respiratory failure
- Following specific diet recommendations and maintaining a specific body weight
- Influenza, pneumococcal, pertussis, and COVID-19 vaccinations
- Maintaining blood pressure within target range
- Limiting salt (sodium) intake
- Following medical practitioner recommendations for exercise and recognizing activities such as walking or gardening may be helpful
- Knowledge of self-management and written action plan that identifies HF triggers and management of worsening symptoms (e.g., diuretic regimen, fluid restriction for volume overload)
- Performing daily weights and reporting a gain of > 2 lbs/day to medical practitioner
- Recognizing signs of depression or anxiety and seeking emotional support from medical practitioner and community resources (e.g., Together in HF from the American Association of HF Nurses, American Heart Association, and the American Heart Failure Society of America)
- Strict medication adherence
- Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
- When to call medical practitioner for new or worsening symptoms (sudden weight gain of more than 2 pounds in 1 day or 5 pounds in 1 week, or weight gain parameters from medical practitioner, trouble breathing at rest, new or increased swelling of your legs or ankles, swelling or pain in your abdomen, breathing trouble at night, frequent coughing that doesn't go away)
- When to seek emergency care (e.g. severe shortness of breath, severe chest pain that does not resolve with rest or nitroglycerine, pink, foamy mucus with cough and shortness of breath, rapid or irregular heartbeat, passing out or fainting, etc.)
- Discharge medications
 - Aldosterone Antagonists
 - Angiotensin-Converting Enzyme (ACE)
 - Angiotensin II Receptor Blockers (or Inhibitors)
 - Angiotensin-Receptor Neprilysin Inhibitors (ARNIs)
 - Beta Blockers
 - Diuretics
 - Hydralazine and isosorbide dinitrate
- Osteomyelitis
 - Follow-up appointments
 - Primary medical practitioner
 - Orthopedic surgeon
 - Referrals and consults
 - Home care for ongoing skilled needs
 - Smoking cessation counseling
 - Patient Education
 - Activity restrictions (e.g., not driving or operating machinery) as recommended by medical practitioner or pharmacist, especially when taking pain medicines or muscle relaxants
 - Care of the affected extremity (e.g., keeping the area free from injury, use of a splint, sling, or brace as directed by medical practitioner)
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations
 - Proper care of catheter site, if indicated (e.g., keeping site clean and dry, monitoring for infection)
 - Proper wound care, if wound present (e.g., keeping wound clean, dressing wound and washing of hands when changing dressings, etc.)
 - Recognizing factors or conditions that increase risk for infection and worsening symptoms (e.g., prolonged immobility, malnutrition, etc.)
 - Smoking cessation, if indicated
 - Strict medication adherence
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)

- When to call medical practitioner for new or worsening symptoms (increasing pain, swelling, warmth, redness or drainage from the infected area, more pain around the infected area, fever)
- Discharge medications
 - Analgesic
 - Antifungal
 - Anti-infective
- Pneumonia
 - Follow-up appointments
 - Primary medical practitioner
 - Pulmonologist
 - Referrals and consults
 - Home care for patients with ongoing skilled needs
 - Consider palliative care or end of life support for advanced conditions contributing to pneumonia
 - Smoking cessation counseling
 - Patient Education
 - Breathing exercises (e.g., incentive spirometry, rhythmic inhalation and coughing)
 - Drinking liquids to stay hydrated to promote coughing up mucus (unless told to limit fluids)
 - Influenza, pneumococcal, pertussis and COVID-19 vaccinations ⁽⁷⁶⁾
 - Recognizing factors or conditions that increase risk for pneumonia or worsen symptoms (e.g., smoking and air pollution, upper respiratory infections, prolonged immobility, malnutrition, poor handwashing, etc) and
 - Following instructions for oxygen therapy, if appropriate for chronic respiratory failure
 - Medication adherence
 - Smoking cessation, if indicated ⁽⁷⁸⁾
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - Use of a cool mist humidifier to increase air moisture
 - When to call medical practitioner for new or worsening symptoms (e.g., shortness of breath, chest pain when you take a breath, wheezing, coughing up mucus that is thick or blood-stained, fever higher than 101.5 F (38.6 C))
 - Discharge medications
 - Analgesic
 - Anti-infective, anti-viral, or anti-fungal
 - Inhaled glucocorticosteroids
 - Mucolytic Agents
 - Oral glucocorticosteroids
- Pyelonephritis or complex UTI
 - Follow-up appointments
 - Primary medical practitioner
 - Urologist
 - Referrals and consults
 - Home care for ongoing skilled needs
 - Smoking cessation counseling
 - Patient Education
 - Influenza, pneumococcal, pertussis and COVID-19 vaccinations
 - Knowledge of factors or conditions that increase risk for infection and worsen symptoms (e.g., prolonged immobility, malnutrition, etc.)
 - Limiting alcohol intake and drinking enough fluids to keep urine light yellow in color (unless told to limit fluids)
 - Medication adherence and completing full antibiotic course
 - Preventing future infection (keeping genital area clean, proper wiping of genital for woman, urinating frequently, practicing safe sex)
 - Proper care of indwelling catheters, if applicable (including clean procedures and when to replace)
 - Smoking cessation, if indicated

- Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
- When to call health care medical practitioner for new or worsening symptoms (fever, shaking chills, vomiting, urinating very little or not at all, blood in your urine, dark-colored or foul-smelling urine, side or back pain that gets worse)
- Discharge medications
 - Analgesic
 - Anti-infective
- Respiratory failure
 - Follow-up appointments
 - Primary medical practitioner
 - Pulmonologist
 - Referrals and consults
 - Enroll patient in call back program, if available ⁽⁶⁶⁾
 - Palliative care or end of life support for advanced lung disease
 - Refer patient to a medical home, disease management program, or case management program, if appropriate and available ^(67, 68)
 - Nutritional counseling or supplementation, if severely overweight or malnourished
 - Smoking cessation counseling
 - Patient Education
 - Avoidance of air pollutants, including a smoking cessation program, if indicated ⁽⁶⁹⁾
 - Breathing exercises to manage shortness of breath (e.g., pursed-lip breathing)
 - Following instructions for oxygen therapy, if appropriate for chronic respiratory failure
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations ⁽⁷³⁾
 - Recognizing signs of depression or anxiety and seeking emotional support from medical practitioner and community resources (e.g., Better Breathing Support Groups)
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - When to call medical practitioner for new or worsening symptoms (e.g., increased swelling in legs, feet or abdomen, wheezing, coughing up blood in sputum, fever)
 - When to seek emergency care (e.g., difficulty breathing, shortness of breath, chest pain)
 - Discharge medications
 - Anticholinergics
 - Beta-Agonists
 - Inhaled glucocorticosteroids
 - Mucolytic Agents
 - Oral glucocorticosteroids (prednisone methylprednisone)
 - Phosphodiesterase -4 Inhibitors
- Sepsis
 - Follow-up appointments
 - Primary medical practitioner
 - Infectious disease specialist
 - Wound care specialist
 - Referrals and consults
 - Home care for ongoing skilled needs
 - Consider palliative care or end of life support for advanced conditions contributing to sepsis
 - Smoking cessation counseling
 - Patient Education
 - Influenza, pneumococcal, pertussis and COVID-19 vaccinations
 - Knowledge of factors or conditions that increase risk for infection and worsen symptoms (e.g., prolonged immobility, malnutrition)
 - Limiting alcohol intake and drinking enough fluids to keep urine light yellow in color, unless told to limit fluids
 - Medication adherence

- Proper wound care, if wound present or post surgery (e.g., keeping wound clean, dressing wound and washing of hands when changing dressings, etc.)
- Smoking cessation, if indicated
- Staying physically active with exercise as instructed by medical practitioner
- Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
- Understanding Post Sepsis Syndrome and long-term effects of sepsis (e.g., insomnia, extreme fatigue, poor concentration, decreased mental or cognitive functioning, loss of self-esteem, nightmares, vivid hallucinations, panic attacks, disabling muscle and joint pain)
- When to call medical practitioner for new or worsening symptoms (chills, sweats, nausea or vomiting, fever higher than 101.5 F (38.6 C), body aches, fast heartbeat, dizziness when you stand up or fainting, confusion, redness, swelling, or drainage of pus from any wounds or from around area where an IV goes into skin)
- Discharge medications
 - Analgesic
 - Anti-infective, anti-viral, or anti-fungal
- Stroke
 - Follow-up appointments
 - Primary medical practitioner
 - Neurologist
 - Referrals and consults
 - Home or community-based rehabilitation or inpatient rehabilitation program
 - Palliative care or end of life support for advanced disease
 - Smoking cessation counseling
 - Patient Education
 - Achieving and maintaining a healthy weight
 - Elimination or reduction of alcohol intake
 - Influenza, pneumococcal, pertussis, and COVID-19 vaccinations
 - Recognizing depression is common after stroke and seeking emotional support
 - Smoking cessation and avoiding passive smoke ⁽⁷⁹⁾
 - Staying physically active as advised by medical practitioner
 - Strict medication adherence ⁽⁸⁰⁾
 - Taking care of health (e.g., getting at least 7h or more per day of sleep, eating a healthy diet, learning ways to manage stress and staying physically active as advised by medical practitioner)
 - Understanding risk of having another stroke and any necessary life-style modification(s) (e.g., maintaining a healthy weight and blood pressure, quitting smoking, and for ischemic stroke healthy eating to control blood sugar and cholesterol level)
 - Understanding risk of falling and changes necessary to maintain a safe living environment if indicated
 - Understanding common problems associated with stroke (e.g., changes in behavior, difficulty doing tasks, memory, moving and doing ADLs, muscle spasms, swallowing, lack of sensation or awareness of one part of the body, thinking, talking, etc.) and any planned therapy services (e.g., physical, occupational or speech therapy)
 - When to call medical practitioner for new or worsening symptoms (e.g., loss of balance or coordination, personality changes, dizziness or lightheadedness, headache that will not go away after treatment, confusion, forgetfulness)
 - When to seek emergency care (e.g., weakness, numbness, drooping, or tingling of face, trouble seeing, severe headache, seizure, trouble thinking, trouble swallowing, trouble talking or understanding speech, feeling dizzy, weakness, numbness or tingling in arm or leg, trouble walking or moving arm or leg, loss of bladder or bowel control)
 - Discharge medications
 - Anticoagulant
 - Antihypertensive
 - Antiplatelet agent
- Assess social determinants of health (SDOH)

● Adult ⁽⁸¹⁾

- Assess for alcohol or substance use
 - Care coordination with outpatient medical or behavioral health providers
 - Provide information about community resources and programs
 - Referral to addiction social support systems
- Assess for domestic abuse, abuse of elderly, and vulnerable adults ⁽⁸²⁾
 - Provide information about legal and community resources
 - Create a long-term safety plan or make appropriate referral
 - Consider intensive home nursing interventions and psychosocial counseling interventions ⁽⁸³⁾
- Assess need for end of life planning
 - Consider palliative care for patients with end stage disease
 - Consider hospice care for terminally ill patients with a life expectancy of less than 6 months ⁽⁸⁴⁾
- Assess for strained or limited financial resources
 - For income insecurity, identify income-support resources ⁽⁸⁵⁾
 - Consider direct income assistance benefits and indirect income assistance programs (i.e., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)
 - Consider referral to Area Agencies on Aging (AAA) for information on nutrition and meals programs, home homemaking services, transportation resources
 - For food insecurity provide information about local food banks and Supplemental Nutrition Assistance Program (SNAP) incentive programs ⁽⁸⁶⁾
- Assess for functional or ADL impairment
 - Consider an alternate level of care
 - Assess the need for a capacity evaluation ⁽⁸⁷⁾
 - Provide information to patient and family about advance directive or power of attorney
- Assess for mental health co-morbidities
 - Arrange for appointment with primary care provider or behavioral health specialist upon discharge
 - Coordinate care with outpatient behavioral health providers
 - Provide information on community resources
- Assess risk for readmission
 - Consider home care services
 - Consider transfer to an alternative level of care or readmission reduction program ⁽⁸⁸⁾
 - Consider telehealth and mobile health interventions
 - Provide patient and family with information on disease management programs
 - For older individuals at risk for malnutrition consider dietitian evaluation, home-delivered meals, and other community resources ⁽⁸⁹⁾
- Assess risk for treatment nonadherence
 - For patients with poor insight into illness or noncompliance with treatment, introduce peer support groups to help improve service user experience and quality of life (National Institute for Health and Care Excellence (NICE)) ⁽⁹⁰⁾
 - Consider in-home intervention to prevent medical relapse, or readmission, and to enhance medication compliance
 - For patients at risk of medication nonadherence, consider patient education and behavioral support initiatives such as text message reminders to take a medication and electronic medication dispensers
- Assess for environmental exposure to toxic substances and other physical hazards
 - For tobacco use and/or exposure, information should be given about tobacco-use cessation and the risk of second-hand smoke ⁽⁹¹⁾
 - For lead poisoning, provide community resources for primary prevention and secondary prevention strategies including blood lead level testing and follow up ⁽⁹²⁾
- Assess for unstable living environment, housing vulnerability or potential for homelessness upon discharge ⁽⁹³⁾
 - Assist or provide health insurance application

- Assist or provide homeless shelter application and provide information to patient
- Consider referral to walk-in state or federally sponsored outpatient programs
- Consider referral to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system ⁽⁹⁴⁾
- For individuals with homelessness and the greatest service need (i.e., serious mental illness, history of hospitalizations, functional impairment) consider Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community-Based Treatment services ⁽⁹⁵⁾
- Assess for unsafe home environment
 - If home environment is unsafe, consider a skilled nursing facility (SNF)
 - For patients at risk for falls, provide information on community-based fall prevention programs through the National Council on Aging
- Pediatric ⁽⁹⁶⁾
 - Assess for alcohol or substance use
 - Coordinate care with outpatient primary care or behavioral health providers
 - Refer to addiction social support systems
 - Provide information about community resources and programs for family-centered treatment (e.g., intensive case management, detox, outpatient, residential treatment, day treatment, and medication assisted treatment)
 - Provide information about home visitation programs (e.g., Parent child Assistance Program (PCAP) for pregnant and parenting women with substance use disorders aftercare) ⁽⁹⁷⁾
 - Provide information about Cognitive Behavioral Therapy (CBT) Multidimensional Family Therapy (MDFT) and/or Pharmacotherapy for youth with co-occurring emotional/mental disorders ⁽⁹⁸⁾
 - Assess for child maltreatment or abuse (e.g., sexual and/or psychological abuse)
 - Arrange for alternative housing situation, foster care, or shelter if home environment is unsafe
 - Assist with creating a long-term safety plan or make appropriate referral
 - Refer to child protective services (CPS) and other domestic violence services
 - Refer for psychosocial support
 - Provide community resources and legal information
 - Provide information on parent training programs such as the Nurse Family Partnership Program (NFP) ⁽⁹⁹⁾
 - Assess need for end of life planning, palliative care, or hospice services
 - Consider palliative care for children with end stage disease
 - Consider hospice care for terminally ill children with a life expectancy of less than 6 months
 - Assess for environmental exposure to toxic substances and other physical hazards
 - For lead poisoning provide community resources for primary prevention and secondary prevention strategies (e.g., blood lead level testing and follow up) ⁽⁹²⁾
 - For tobacco use (e.g., cigarettes, E-cigarettes, vaping) provide education and/or brief counseling about tobacco use cessation and the risk of second-hand smoke ⁽⁹¹⁾
 - Assess for infants and children with special health care needs
 - Provide anticipatory guidance and education to caregivers regarding medical condition and expected progress after discharge over the next few days, months, or years, as appropriate ⁽¹⁰⁰⁾
 - Consider referral for follow-up care to neonatologist or other medical subspecialist for infants or children with ongoing and/or chronic medical issues, (e.g., bronchopulmonary dysplasia or feeding dysfunction) ⁽¹⁰¹⁾
 - Initiate referral for adaptive or assistive medical equipment and equipment for technology-dependent children (e.g., home oxygen, ventilators, monitors, or tube feeding) ⁽¹⁰¹⁾
 - Consider referral for neurodevelopmental assessment for high-risk infants
 - Consider referral to post-discharge follow-up programs ⁽¹⁰²⁾
 - Consider referral for behavioral support and/or counseling for caregivers through therapy groups, telephone support, and/or peer support groups for caregivers of children with similar health needs ⁽¹⁰³⁾
 - Consider referral to lactation consultant in the hospital and home visits for infants at risk for poor lactation or suboptimal infant breast-feeding behavior ⁽¹⁰⁴⁾
 - Consider referral for respite services

- Assess for strained or limited financial resources
 - For food insecurity provide information about local food banks and special nutrition programs (e.g., Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), U.S. Department of Agriculture Summer Food Service ⁽⁸⁶⁾)
 - For income insecurity identify income resources and refer to direct and/or indirect income assistance programs (e.g., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)
- Assess for mental health co-morbidities
 - Arrange appointment with primary care provider or behavioral health specialist upon discharge
 - Coordinate care with outpatient behavioral health providers
 - ⁽¹⁰⁵⁾ Provide information about suicide prevention, including the National Suicide Prevention hotline
 - Consider referral for psychiatric services for suicide risk screening and intervention ⁽¹⁰⁵⁾
 - Consider referral to school-based mental health interventions ⁽¹⁰⁶⁾
 - Provide information about Intensive Community-based treatment (e.g., Functional Family Therapy, Intensive Child and Adolescent and Psychiatric Services, Multidimensional Family Therapy, and Intensive Case Management)
 - Consider referral to day treatment program, residential addiction services, or partial hospitalization
- Assess risk for treatment nonadherence
 - Provide information about educational and behavioral support initiatives for children at risk of medication nonadherence and their caregivers ⁽¹⁰⁷⁾
 - Provide information about peer support groups for children and/or families with poor insight into illness or noncompliance with treatment ⁽¹⁰⁸⁾
 - Consider home care services or community resources to prevent medical relapse, or readmission, and to enhance medication compliance
- Assess risk for readmission
 - Consider transfer to an alternative level of care such as subacute care
 - Consider home care services
 - Consider early childhood home visiting programs (e.g., Early Head Start; Early On; Nurse Family Partnership; and the Maternal, Infant, and Early Childhood Home Visiting Program) ⁽¹⁰²⁾
 - Consider telehealth and web-based training programs (e.g., telepsychiatry or telebehavioral health and mobile health interventions) ⁽¹⁰⁹⁾
 - Consider intensive in-home behavioral health service
 - Provide patient and family with condition-specific information including appropriate planning for prevention of RSV infection and appropriate immunizations if appropriate
- Assess for special health risk behaviors in adolescence and young adulthood
 - Arrange follow-up appointment with primary care provider and/or specialists upon discharge
 - Coordinate care with outpatient behavioral health providers
 - Provide information on mentoring programs, (e.g., Big Brothers Big Sisters of America)
 - Consider referral for Intensive Community-Based treatment for adolescents with comorbid behavioral health issues
 - Provide community resources for teen pregnancy prevention and family planning (use of birth control, subsequent pregnancies) ⁽¹¹⁰⁾
- For eating disorders
 - Provide information on nutrition services including education and resources for diet management, physical activity, and promotion of healthy eating attitudes and behaviors ⁽¹¹¹⁾
 - Provide information on outpatient psychosocial interventions and Family-Based Treatment ⁽¹¹²⁾
 - Refer to an intensive weight management program that includes culturally tailored interventions ⁽¹¹³⁾
- Provide parenting resources
 - Provide information on Family Resource Centers and Parenting support groups for young, first-time parents, (e.g., National parent help line) ⁽¹¹⁴⁾
 - Provide information on community resources to improve parenting skills (face-to-face or by telephone) ⁽¹¹⁵⁾

- Assess for unstable living environment, housing vulnerability or potential for homelessness upon discharge ⁽⁹³⁾
 - Assist or provide health insurance application
 - Assist or provide homeless shelter application to caregivers
 - Consider referral to local housing coordinator or case manager for immediate access to permanent supportive housing and coordinated access system ⁽⁹⁴⁾
 - Consider referral to walk-in state or federally sponsored outpatient programs
 - Consider referral to Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community Treatment services ⁽¹¹⁶⁾
- Plan for follow-up care ^(117, 118)
 - Assist or arrange follow-up care or testing
 - Solicit input from patient or caregiver about convenient dates, times, and places
 - Coordinate appointments with practitioners, testing sites, and other service areas
 - Organize post discharge services and medical supplies or equipment
 - Ensure patient or caregiver understands the importance of services
 - Discuss what to expect with each service
 - Clarify any questions about medical care and follow-up
 - Refer patient to a medical home, disease management program, or case management program, if appropriate and available ^(67, 68)
 - Enroll patient in call back program, if available ⁽⁶⁶⁾
- Review discharge instructions ⁽¹¹⁹⁾
 - Identify any pending in-hospital tests or studies and determine who will follow-up with them ⁽¹²⁰⁾
 - Educate patient or caregiver about relevant diagnoses ⁽¹²¹⁾
 - Review contact information for medical practitioner, specialty care, services, and outpatient testing
 - Provide a written list of scheduled appointments and tests
 - Assess understanding of plan by asking patient to explain it in his or her own words (may require discussion with caregiver) ^(122, 123)
- Medication reconciliation ⁽⁴⁶⁾
 - Provide list and discuss
 - How and when medications should be taken
 - Reasons for medications
 - Common side effects
 - Changes to previously prescribed medications
 - Review of over-the-counter (OTC) medications, supplements, and herbal remedies
 - Assess for potential drug interactions or drug allergies
 - Assess the risks and benefits of opiate therapy for acute/chronic pain management, following recommendations for initiation, continuation, and follow-up management of opioid therapy ⁽¹²⁴⁾
 - Address potential reasons for non-adherence ⁽¹²⁵⁾
 - Does the patient have a mechanism for filling prescriptions?
 - Does the patient have the means to pay for medications?
 - Are there community resources or discount programs available?
 - Are there generic alternatives?
 - Ensure prescriptions are provided or e-prescriptions sent
 - Ensure patient or caregiver verbalizes understanding of regimen ⁽¹²²⁾
- Assess understanding of when and where to seek help ⁽¹²²⁾
 - What constitutes an emergency
 - What to do in the case of an emergency
 - When to contact medical practitioner
 - Who to call on weekends or evenings
- Identify and address transportation needs
- Obtain and complete forms for home care agency
- Obtain discharge summary ⁽¹²⁶⁾
 - Transmit to agency

- Transmit to medical practitioner prior to the first follow-up visit ⁽¹²⁷⁾

● **SKILLED NURSING FACILITY (SNF)**

● **Home setting inappropriate**

- Outpatient or home care services unavailable or inappropriate due to clinical complexity

● **Patient or caregiver unable to manage care**

● **Cognitive limitation**

- Communication deficit prevents learning care tasks
- Memory deficit prevents managing care tasks
- Perception or processing deficit

● **Physical limitation**

- Physically unable to render care (e.g., wound location, contractures, unable to lift patient, has obesity)
- Lacks dexterity, motor strength, or skills required to manage care
- Comorbidity prevents management of care (e.g., blindness, paralysis)

● **Skilled service required for assessment, treatment, monitoring, or education**

- Medical practitioner oversight at least 1 time per week
- Nursing daily or skilled therapy 1-2h per day at least 5d per week

● **SNF available**

- Provide information to patient or caregiver
- Review site information or visit site ⁽¹²⁹⁾

● **Complete prior to facility transfer**

- Medication reconciliation ^(46, 124)
- Obtain and complete forms for facility
- **Obtain discharge summary** ⁽¹²⁶⁾
 - Transmit to facility
 - Transmit to medical practitioner ⁽¹²⁷⁾
- Arrange transportation

● **SUBACUTE CARE (SAC)**

● **Lower level of care inappropriate**

- Outpatient or home care services unavailable or inappropriate due to clinical complexity

● **Patient or caregiver unable to manage care:**

● **Cognitive limitation**

- Communication deficit prevents learning care tasks
- Memory deficit prevents managing care tasks
- Perception or processing deficit

● **Physical limitation**

- Physically unable to render care (e.g., wound location, contractures, unable to lift patient, has obesity)
- Lacks dexterity, motor strength, or skills required to manage care
- Comorbidity prevents management of care (e.g., blindness, paralysis)

● **Skilled service required for assessment, treatment, monitoring, or education**

- Medical practitioner oversight at least 2 times per week
- Nursing 4h or more per day or skilled therapy 2-3h per day at least 5d per week

● **SAC facility available**

- Provide information to patient or caregiver
- Review site information or visit site ⁽¹²⁹⁾

● **Complete prior to facility transfer**

- Medication reconciliation ^(46, 124)
- Obtain and complete forms for facility
- **Obtain discharge summary** ⁽¹²⁶⁾
 - Transmit to facility
 - Transmit to medical practitioner ⁽¹²⁷⁾
- Arrange transportation

● **LONG-TERM ACUTE CARE (LTAC)**

- **Lower level of care inappropriate due to clinical complexity**

- **Skilled service required for assessment, treatment, monitoring, or education**
 - Medical practitioner assessment or intervention daily
 - Nursing at least 6.5h per day
- **LTAC facility available**
 - Provide information to patient or caregiver
 - Review site information or visit site ⁽¹²⁹⁾
- **Complete prior to facility transfer**
 - Medication reconciliation ^(46, 124)
 - Obtain and complete forms for facility
 - **Obtain discharge summary** ⁽¹²⁶⁾
 - Transmit to facility
 - Transmit to medical practitioner ⁽¹²⁷⁾
 - Arrange transportation
- **ACUTE REHABILITATION**
 - **Lower level of care inappropriate due to clinical complexity**
 - Requires comprehensive rehabilitation program 3h or more per day at least 5d per week
 - Rehabilitation medical practitioner assessment or oversight at least 3 times per week
 - Nursing for daily assessment, treatment, monitoring, or education
 - **Acute rehab facility available**
 - Provide information to patient or caregiver
 - Review site information or visit site ⁽¹²⁹⁾
 - **Complete prior to facility transfer**
 - Medication reconciliation ^(46, 124)
 - Obtain and complete forms for facility
 - **Obtain discharge summary** ⁽¹²⁶⁾
 - Transmit to facility
 - Transmit to medical practitioner ⁽¹²⁷⁾
 - Arrange transportation

Notes:**1:**

Major depression can play a significant role in readmission of patients with advanced chronic diseases such as heart failure, chronic obstructive pulmonary disease, and coronary artery disease (post myocardial infarction) (Almagro et al., *Respiration* 2006; 73(3): 311-317; Parashar et al., *Arch Intern Med* 2006; 166(18): 2035-2043; Jiang et al., *Arch Intern Med* 2001; 161(15): 1849-1856). Depression affects almost one third of medical inpatients up to 30 days post-discharge, with depressed patients with short-term poor prognoses having a 73 percent risk for readmission within 30 days compared to patients without depressive symptoms and similar findings at 90 days (Pederson et al., *J Hosp Med* 2016, 11: 373-80). In 2013, mood disorders had an all-cause 30-day hospital readmission rate of 15.3 (Fingar, *Trends in Hospital Readmissions for Four High-Volume Conditions, 2009-2013. HCUP Statistical Brief #196. 2015*).

2:

Schizophrenia and other psychotic disorders have been ranked by the Agency for Healthcare Research and Quality (AHRQ) as having a high-volume rate of 22.7 for all-cause readmissions within 30 days (Fingar, *Trends in Hospital Readmissions for Four High-Volume Conditions, 2009-2013. HCUP Statistical Brief #196. 2015*).

3:

Reference: (Maestri et al., *Psychopharmacol Bull* 2018, 48: 8-15; Taipale et al., *Schizophr Bull* 2018, 44: 1381-7)

4:

Inadequate functional health literacy has been linked to increased use of emergency department services, a higher risk of hospitalization, rehospitalization, and higher overall health care costs (Kripalani et al., *JAMA* 2007; 297(8): 831-841). Independent of education level, patients with low health literacy are likely to struggle with reading comprehension of prescription bottles, appointment slips, self-care instructions, and health education brochures. Written materials in some hospitals are written at third or fourth grade reading level using the patient's native language. They include the reason for hospitalization, symptoms to watch out for, a plain language list of appointment follow-ups, and information related to medications (Boutwell, et al., *AHRQ Designing and Delivering Whole-Person Transitional Care: The Hospital Guide to Reducing Medicaid Readmissions. September 2016*). Similarly, patients with chronic diseases, who struggle with inadequate literacy, are less likely to know basic elements of how to care for their medical problems (Baker et al., *Am J Public Health* 2002; 92(8): 1278-1283; Mitchell et al., *J Health Commun* 2012, 17 Suppl 3: 325-38). Interventions that focus on increasing self-care competence demonstrate reduced readmission rates (Leppin et al., *JAMA Intern Med* 2014, 174: 1095-107).

5:

Of all Medicare patients discharged from the hospital with acute myocardial infarction, congestive heart failure, or pneumonia, Black patients had higher 30-day readmission rates than their White counterparts. It was further noted that the difference was related, in part, to higher readmission rates among hospitals that have a disproportionate Black patient population (Joynt et al., *JAMA* 2011, 305: 675-81). Racial disparity is an area that requires further research.

6:

From July 2012 to June 2015, the Centers for Medicare and Medicaid's national hospital compare for all-cause 30-day readmission rate for an acute myocardial infarction (AMI) was 16.8 (Centers for Medicare & Medicaid Services, *Hospital Compare datasets. 2016*). Patients who have had a myocardial infarction (MI) with comorbid chronic obstructive airway disease (COPD) are at a higher risk for rehospitalization during the post-MI period. They also report a poorer quality of life, decreased physical functioning, and have a greater prevalence of angina one year post-MI, compared to those patients who have had an MI and who do not have comorbid COPD (Salisbury et al., *Am J Cardiol* 2007; 99(5): 636-641).

7:

Reference: (Kini et al., *Circ Cardiovasc Qual Outcomes* 2018, 11: e004788; Martin et al., *J Am Heart Assoc* 2018, 7: e009339; Shah et al., *Circ Heart Fail* 2018, 11: e004310)

8:

Reference: (Kini et al., Circ Cardiovasc Qual Outcomes 2018, 11: e004788)

9:

Reference: (Martin et al., J Am Heart Assoc 2018, 7: e009339)

10:

Reference: (Nguyen et al., J Am Heart Assoc 2018, 7;; Shah et al., Circ Heart Fail 2018, 11: e004310; Mahmoud et al., J Am Heart Assoc 2018, 7:)

11:

Reference: (Nguyen et al., J Am Heart Assoc 2018, 7:)

12:

Reference: (Shah et al., Circ Heart Fail 2018, 11: e004310)

13:

Reference: (Smilowitz et al., Circulation 2018, 137: 2332-9; Shah et al., Circ Heart Fail 2018, 11: e004310; Mahmoud et al., J Am Heart Assoc 2018, 7:)

14:

The Centers for Medicare and Medicaid reports patients admitted with coronary artery bypass graft surgery (CABG) experienced a 30-day all-cause readmission rate of 14.4 from July 2012 to June 2015, with a 3.2 mortality rate in that same period (Centers for Medicare & Medicaid Services, Hospital Compare datasets. 2016). In patients deemed low risk for post coronary artery bypass graft mortality, the presence of diabetes, a length of stay longer than 5 days, or the need for blood product transfusions were associated with an increased risk for readmission (Sun et al., Heart Surg Forum 2008; 11(6): E327-332).

15:

Out of the four high-volume conditions (acute myocardial infarction, heart failure, chronic obstructive lung disease, and pneumonia), heart failure was ranked as the condition with the highest 30-day all-cause readmission from July 2012 to June 2015 by the Centers for Medicare and Medicaid. This condition had a readmission rate of 21.9, with a 12.1 mortality rate in this same period (Centers for Medicare & Medicaid Services, Hospital Compare datasets. 2016).

16:

Comprehensive discharge planning along with post discharge support may reduce readmission rates and improve health outcomes for patients with heart failure (HF). Post discharge support shown to be of benefit includes a single home visit, home visit with frequent follow-up phone calls, extended home services, and day hospital services (Phillips et al., JAMA 2004; 291(11): 1358-1367; Bueno et al., JAMA 2010, 303: 2141-7; Feltner et al., Ann Intern Med 2014, 160: 774-84). Additionally, in a study evaluating heart failure patients treated and released from the emergency department, outcomes improved for those patients receiving follow-up care within the first 30 days of discharge provided by both their primary care physician and a cardiologist. This was attributed to the use of diagnostic interventions and medication management as a result of physician collaboration (Lee et al., Circulation 2010, 122: 1806-14).

17:

For patients admitted with heart failure (HF), post discharge assessment is now considered an essential part of their aftercare plan with early follow-up. Initial studies have shown post discharge assessment is associated with lower rates of all-cause readmission (Metra et al., Circulation 2010, 122: 1782-5).

18:

Patients who are younger than age 65, at the time of initial heart failure hospitalization, are at higher risk for readmission (Aranda et al., Clin Cardiol 2009; 32(1): 47-52).

19:

Heart failure patients with a high pre-discharge BNP that has not decreased by more than 50 percent of the admission value are at a higher risk for readmission (Cournot et al., Am Heart J 2008; 155(6): 986-991; Logeart et al., J Am Coll Cardiol 2004; 43(4): 635-641).

20:

Patients with clinically stable heart failure, regardless of ejection fraction, are at higher risk for readmission after undergoing elective major noncardiac surgery (Hammill et al., Anesthesiology 2008; 108(4): 559-567; Xu-Cai et al., Mayo Clin Proc 2008; 83(3): 280-288).

21:

Worsening renal function is a change in serum creatinine greater than, or equal to 0.3 mg/dL from admission to discharge (Patel et al., Am Heart J 2010; 160(1): 132-138 e131).

22:

Reference: (Shah et al., Clin Cardiol 2018, 41: 916-23)

23:

Reference: (Al-Omary et al., Heart Lung Circ 2018, 27: 917-27; Sokoreli et al., Eur J Heart Fail 2018, 20: 689-96)

24:

Reference: (Chen et al., Medicine (Baltimore) 2018, 97: e9629)

25:

(Sokoreli et al., Eur J Heart Fail 2018, 20: 689-96)

26:

The Centers for Medicare and Medicaid reports patients admitted with chronic obstructive pulmonary disease (COPD) experienced a 30-day all-cause readmission rate of 20 from 2012 to 2015, with a mortality rate of 8 in that same time period (Centers for Medicare & Medicaid Services, Hospital Compare datasets. 2016). One out of five patients discharged with the diagnosis of COPD require rehospitalization within 30 days. Research shows early follow-up post-hospitalization by either the patient's primary care physician or pulmonologist can result in decreased emergency room visits and hospital readmission within 30 days of discharge (Shah et al., Chest 2016, 150: 916-26).

27:

Patients admitted with chronic obstructive pulmonary disease (COPD) within the past year are at an increased risk of readmission. Evidence demonstrates a marked increase in the risk of readmission with each new exacerbation requiring hospitalization. The first three months following a severe exacerbation is considered a high risk period for readmission (Echevarria et al., Thorax 2017, 72: 686-93). Additional risk factors include hypercapnia (P_{CO_2} greater than 45.0 mmHg) at discharge, long-term oxygen therapy, oral corticosteroid use, and poor health-related quality of life, and lack of routine physical activity. Post discharge home care including disease education, telephone support, and rapid access to outpatient clinics may reduce readmission rates in COPD patients (Shah et al., Chest 2016, 150: 916-26).

28:

Health-related quality of life is a comprehensive measure of health status. Recommended measurement tools include the COPD Assessment Test (CAT) and the COPD Control Questionnaire (CCQ) (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; Zhou et al., Respiriology 2017, 22: 251-62).

29:

Reference: (Abdulai et al., Am J Respir Crit Care Med 2018, 197: 433-49; Ehsani et al., Int J Chron Obstruct Pulmon Dis 2019, 14: 39-49)

30:

Reference:(Spece et al., Ann Am Thorac Soc 2018, 15: 1033-8; Bottle et al., In: Factors associated with hospital emergency readmission and mortality rates in patients with heart failure or chronic obstructive pulmonary disease: a national observational study. 2018)

31:

Reference: (Singh et al., J Allergy Clin Immunol Pract 2019, 7: 279-85 e6)

32:

Reference: (Bottle et al., In: Factors associated with hospital emergency readmission and mortality rates in patients with heart failure or chronic obstructive pulmonary disease: a national observational study. 2018)

33:

Reference:(Criner et al., Chest 2018, 153: 1497-8)

34:

Reference: (Cerezo Lajas et al., Lung 2018, 196: 185-93)

35:

From July 2012 to June 2015, the Centers for Medicare and Medicaid's national hospital compare all-cause 30-day readmission rate for pneumonia was 17.1, with a 16.3 mortality rate (Centers for Medicare & Medicaid Services, Hospital Compare datasets. 2016). Pneumonia patients at highest risk for readmission include those who are age 65 or older, those who experience worsening of comorbidities during hospitalization, or have unstable vital signs at discharge. While these risk factors may have a poor predictive value, it is important to note that the majority of pneumonia readmissions are related to comorbid conditions rather than complications of pneumonia. Over 40 percent of readmissions occur within a week of initial discharge. The most frequently identified comorbidities associated with readmission include coronary artery disease, chronic obstructive pulmonary disease and neurological diseases (Jasti et al., Clin Infect Dis 2008; 46(4): 550-556). Careful attention to the clinical stability of patients with comorbid conditions at and following hospital discharge may decrease the frequency of readmission in these patients (Capelastegui et al., Chest 2009; 136(4): 1079-1085; Jasti et al., Clin Infect Dis 2008; 46(4): 550-556).

36:

The risk of readmission is increased in patients with pneumonia who experience in-hospital treatment failure and one or more instability factors. Treatment failure includes clinical deterioration represented by any of the following: hemodynamic instability, respiratory failure, required mechanical ventilation, or the appearance of new infectious foci. Instability factors include (Capelastegui et al., Chest 2009; 136(4): 1079-1085):

- Temperature > 99.5°F
- Systolic BP < 90 mmHg
- Respiratory rate > 24 breaths/min
- Heart rate > 100 beats/min
- Oxygen saturation < 90% (0.90)

37:

Reference: (Toledo et al., BMJ Open 2018, 8: e020243)

38:

In a large retrospective cohort study, one in three patients with sickle cell disease required hospital readmission within 30 days, a higher rate of readmission when compared with patients hospitalized with asthma, heart failure, pneumonia, and diabetes. Of patients readmitted within 30 days, two-thirds were readmitted within 14 days.

Readmission was most common among patients 18 to 30 years old. Prevention of re-hospitalization may require evaluation and intervention while hospitalized and access to high quality, comprehensive outpatient care in the community setting (Brousseau et al., JAMA 2010, 303: 1288-94).

39:

From July 2012 to June 2015, the Centers for Medicare and Medicaid's national hospital compare all-cause 30-day readmission rate for stroke was 12.5, with a 14.9 mortality rate in that same period (Centers for Medicare & Medicaid Services, Hospital Compare datasets. 2016). There is a high risk of early and long-term hospital readmissions following a transient ischemic attack (TIA) and acute ischemic stroke. Following a TIA/acute stroke, 17.4 percent of patients will require readmission within the first month, and 42.5 percent within the first year. Reasons for readmission include: recurrent stroke, infection, and coronary artery disease. Emphasizing interventions to prevent pneumonia such as influenza and pneumococcal vaccination and improving oral hygiene can improve long term outcomes in stroke patients (Zhong et al., Neurological Sciences 2016: 1-8; Bravata et al., Stroke 2007, 38: 1899-904).

40:

Reference: (Boehme et al., Stroke 2018, 49: 2999-3005)

41:

Reference: (Elgendy et al., JACC Cardiovasc Interv 2018, 11: 2414-24; Echouffo-Tcheugui et al., Eur Heart J 2018, 39: 2376-86; Jin et al., Int J Stroke 2018, 13: 576-84)

42:

Reference: (Hirayama et al., Int J Stroke 2018, 13: 717-24)

43:

From July 2012 to June 2015, the Centers for Medicare and Medicaid's national hospital database rate for all-cause readmissions after hospital discharges was 15.6 (Centers for Medicare & Medicaid Services, Hospital Compare datasets. 2016). Several factors present at hospital admission have been identified as increasing the risk of readmission: 5 or more chronic comorbid conditions, history of depression, and previous hospital admission within the last 30 days (Librero et al., J Clin Epidemiol 1999; 52(3): 171-179; Marcantonio et al., Am J Med 1999; 107(1): 13-17). To prevent or minimize readmissions, key activities may include (Boutwell, et al., AHRQ Designing and Delivering Whole-Person Transitional Care: The Hospital Guide to Reducing Medicaid Readmissions. September 2016):

- Identifying high risk populations based on data collection and analysis
- Approaching population's social and whole-person transitional needs
- Communicating post hospital discharge plans
- Connecting patients to post-acute services when applicable
- Providing real-time information to providers, service agencies, and insurers who will be receiving patients post-discharge
- Timely post discharge contact with patients

44:

Comorbidity and adverse drug reactions in patients greater than 60 years of age have been shown to increase the risk of readmission. Risk reduction may be achieved by careful and frequent monitoring of prescribed drugs in older adults with comorbidities (Zhang et al., BMJ 2009; 338: a2752).

45:

Thirty percent of hospital admissions in older patients are medication related. Patients taking multiple or high risk medications have been shown to benefit from medication reconciliation and transition planning. High risk medications include digoxin, diuretics, anticoagulants, sedatives, opioids, asthma or chronic obstructive pulmonary disease medications, angiotensin converting enzyme inhibitors, or angiotensin receptor blockers. Interventions, such as telephonic follow-up by a pharmacist within 2 to 4 days after discharge or warfarin genotyping, have been shown to decrease risk of readmission (Society of Hospital Medicine, Project BOOST. 2008 [cited Nov 2015]; Fick et al., Arch Intern Med 2003, 163: 2716-24; Epstein et al., J Am Coll Cardiol 2010, 55: 2804-12; Jack et al., Ann Intern Med

2009; 150(3): 178-187).

46:

A comprehensive medication reconciliation process can reduce the likelihood of medication errors, and improve consistency and communication throughout the continuum of care (Medication Reconciliation. Quality and Patient Safety Division. Commonwealth of Massachusetts; 2015). Patients with complex care needs frequently require care in different settings and may be more likely to experience medication problems at each care transition or "handoff" between practitioners (Pediatrics 2008; 122(5): 1119-1126; Coleman et al., Arch Intern Med 2005; 165(16): 1842-1847). When patients move from one setting to another, medication reconciliation ensures that an accurate list of medication is maintained and consistent with the plan of care. Medication reconciliation is a subject of the 2005 National Patient Safety Goals from The Joint Commission (Arora and Farnan, Med Clin North Am 2008; 92(2): 315-324, viii). Additionally, organizations such as the National Transitions of Care Coalition developed a resource document with pertinent elements that they recommend are addressed during the medication reconciliation process (National Transitions of Care Coalition Medication Reconciliation Elements. 2010). Focused attention should be made for those patients at high risk for medication errors (Medication Reconciliation. Quality and Patient Safety Division. Commonwealth of Massachusetts; 2015):

- Low health literacy
- Older patients
- Multiple co-morbidities
- Cognitive impairment due to delirium, medication, and acute illness
- Transfer from facility of the hospital system
- Lack of access to preadmission medication sources
- Provider concern regarding medication safety

47:

Hospitalized patients who are 75 years or older and have a previous hospital admission, comorbidity, functional disability, or prolonged length of stay may be at increased risk for readmission (Garcia-Perez et al., QJM 2011, 104: 639-51).

48:

Post surgical patients who are at high risk for readmission within 30 days include those with multiple medical comorbidities admitted 24 hours prior to surgery, those who develop post surgical complications, and those who undergo complex gastrointestinal procedures. Comorbidities such as diabetes, smoking, and immunosuppression increase the risk for delayed wound healing, surgical site infection, and other postoperative complications. Post surgical complications, such as blood transfusion within 72 hours after surgery may increase the risk of readmission. Patients who undergo complex gastrointestinal surgery (e.g., pancreatectomy, hepatectomy, or colectomy) requiring pre-operative with admission and having pre-operative length of stay longer than 24 hours are considered to be at high risk for readmission (Kassin et al., J Am Coll Surg 2012, 215: 322-30). For high risk surgeries, such as open thoracic aortic aneurysm repair, follow-up with a primary care provider within 30 days after discharge is associated with a reduced 30-day readmission rate (Brooke et al., JAMA Surg 2014, 149: 821-8).

49:

From July 2012 to June 2015, the Centers for Medicare and Medicaid's national hospital compare all-cause 30-day readmission rate for hip/knee replacement surgery was 4.6 (Centers for Medicare & Medicaid Services, Hospital Compare datasets. 2016).

50:

In 2012, the Centers for Medicare and Medicaid Services (CMS) finalized the initial phase of the Readmissions Reduction Program. The program will significantly reduce payments to hospitals with excessive readmissions for beneficiaries with the following admission diagnoses: acute myocardial infarction; acute exacerbation of chronic obstructive pulmonary disease; heart failure; pneumonia, including aspiration pneumonia and sepsis coded with pneumonia present on admission (but not including severe sepsis); and total hip or knee arthroplasty (Centers for Medicare & Medicaid Services, Readmissions Reduction Program. 2018; Centers for Medicare & Medicaid Services, Readmissions Reduction Program. 2016).

51:

Readmissions make up a substantial portion of admissions in the asthma population, with 15 to 30 percent of children with asthma readmitted within a year. This rate varies across hospitals, with racial/ethnic, economic, caregiver health literacy and education, and primary care access disparities contributing to this variation (Auger et al., J Pediatr 2015, 166: 101-8). Black patients, patients with public insurance, adolescents, patients with complex chronic conditions, and patients who have already been admitted within the previous year for asthma had higher rates of readmission to hospitals (Kenyon et al., J Pediatr 2014, 164: 300-5). For those patients with a history of asthma admission, risk of admission increases with each subsequent readmission (Bloomberg et al., Am J Respir Crit Care Med 2003; 167(8): 1068-1076). Interventions aimed at coaching and increasing health literacy have been shown to decrease the risk of readmission (Fisher et al., Arch Pediatr Adolesc Med 2009; 163(3): 225-232). An inpatient focus on medication adherence is key to preventing hospital readmissions (Auger et al., J Pediatr 2015, 166: 101-8).

52:

The following factors have been associated with an increased risk of readmission among children with sickle cell disease (Sobota et al., Pediatr Blood Cancer 2012, 58: 61-5; Sobota et al., Am J Hematol 2010, 85: 24-8; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Strouse et al., Pediatr Blood Cancer 2008, 50: 1006-12):

- History of asthma
- Oxygen requirement during hospitalization and discharge less than 24 hours after transition to room air
- History of greater than 3 hospital admission within the previous 12 months
- Corticosteroid administration

As with many other chronic conditions, follow-up care within 30 days of discharge has been shown to reduce the risk of readmission (Brousseau et al., JAMA 2010, 303: 1288-94; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485). In addition, institution of a standardized pain treatment plan is associated with reduced readmission rates (Sobota et al., Pediatr Blood Cancer 2012, 58: 61-5).

53:

Infants discharged from the neonatal intensive care unit with a history of bronchopulmonary dysplasia are at a higher risk of readmission (Smith et al., J Pediatr 2004; 144(6): 799-803).

54:

Extreme low birth weight infants are at high risk for rehospitalization during their first year of life (Ambalavanan et al., Pediatrics 2011, 128: e1216-25).

55:

Children with severe sepsis at an increased risk of readmission include those with the following (Czaja et al., Pediatrics 2009; 123(3): 849-857):

- Age less than 1 year old
- Neurologic or hematologic dysfunction during the admission
- Cardiovascular or bloodstream infections
- Certain comorbid conditions (e.g., oncologic, hematologic, and other congenital or chromosomal abnormalities)

56:

A length of stay greater than 14 days has been shown to increase risk of readmission in patients with congenital heart disease (Mackie et al., Am Heart J 2008; 155(3): 577-584).

57:

A study of 37 tertiary care pediatric hospitals in the United States indicates a small percentage of patients account for recurrent readmissions. As readmissions increased from 0 to 4 over a 365-day interval, the readmissions were for complex chronic conditions, (e.g., renal, cardiovascular, malignancy, congenital), as opposed to an ambulatory care-sensitive condition (e.g., asthma). Although further strategies are needed to determine avoidable readmissions from admissions that are indicative of quality care, it is noted that the pediatric patients identified in this study may benefit from comprehensive discharge planning in order to ensure a safe transition from the inpatient setting (Berry et al., JAMA 2011, 305: 682-90).

58:

Children with chronic physical or developmental abnormalities who require more intensive health care and related services are considered to be children with medical complexity (CMC). CMC accounts for approximately one-third of child health expenditures with about 80 percent of this cost in hospital care (Coller et al., Pediatrics 2014, 134: e1628-47). CMC is defined by four cardinal characteristics:

- Chronic health condition(s) associated with a high degree of morbidity and/or mortality
- Intense medical needs, including multiple specialty services and care coordination
- Significant functional limitations
- High health care utilization

Once hospitalized, medically complex children typically have longer lengths of stay due to delayed response to treatment when compared to children without disabilities. Following discharge, this patient population is considered high risk for hospital readmission (Neff et al., Acad Pediatr 2015, 15: 191-6; Russell and Simon, Pediatr Ann 2014, 43: e157-62; Simon et al., Pediatrics 2014, 133: e1647-54; Cohen et al., Pediatrics 2012, 130: e1463-70; Berry et al., JAMA 2011, 305: 682-90; Burns et al., Pediatrics 2010, 126: 638-46; Simon et al., Pediatrics 2010, 126: 647-55; Srivastava et al., Pediatr Clin North Am 2005, 52: 1165-87, x).

59:

Hospital readmission is common in extracorporeal membrane oxygenation survivors, who often suffer from significant long-term neurologic and respiratory co-morbidities. Readmission rates are 60 percent in non-neonatal patients without congenital heart disease who are treated with ECMO (Jen and Shew, Pediatrics 2010, 125: 1217-23).

60:

For most patients with acute coronary syndrome, medications used to control ischemia during the hospitalization should be continued following discharge. The medication regimen should be individualized based on diagnostic findings, risk factors, medication tolerance, and procedural interventions (Arnett et al., J Am Coll Cardiol 2019, 74: 1376-414).

61:

Physical activity can reduce symptoms in cardiovascular disease as well as improve other cardiac risk factors by aiding weight loss and weight management. Patients should be encouraged to take daily walks immediately following discharge. In patients with uncomplicated acute coronary syndrome, sexual activity can be resumed within 7 to 10 days and driving can begin 7 days after discharge. In patients who have had a complicated recovery, these determinations should be made on a case-by-case basis. Enrollment in a cardiac rehabilitation program after discharge may enhance blood pressure control, exercise capacity, patient education, compliance and encourage participation in a regular exercise program (Arnett et al., J Am Coll Cardiol 2019, 74: 1376-414). The data are mixed as to whether center-based cardiac rehabilitation increases physical activity; home-based cardiac rehabilitation may lead to increased physical activity (Ter Hoeve et al., Phys Ther 2015, 95: 167-79). Patients who attend early, short-term, comprehensive cardiac rehabilitation programs may have improved prognoses (Rauch et al., Eur J Prev Cardiol 2014, 21: 1060-9).

62:

Prior to discharge from the hospital, patients with acute coronary syndrome should be informed about symptoms of worsening ischemia, and should be instructed when and where they should seek emergency care when such symptoms occur. Patients should be instructed that anginal discomfort lasting longer than two or three minutes requires action. The patient should discontinue physical activity, or remove themselves from the stressful situation. Pain that does not subside immediately should be treated with sublingual nitroglycerin up to three doses or as prescribed. The EMS system should be activated for pain that is not relieved after the first dose of sublingual nitroglycerin (Arnett et al., J Am Coll Cardiol 2019, 74: 1376-414; Anderson et al., J Am Coll Cardiol 2013, 61: e179-347).

63:

The occurrence of depression and anxiety are common following myocardial infarction and can be associated with an increase in behaviors that further increase cardiovascular risk (e.g., smoking, alcohol consumption, over-eating, medication non-adherence) (Berg et al., Journal of psychosomatic research 2018, 112: 66-72). Upon discharge

patients should be made aware of the warning signs of depression and anxiety and the importance of seeking out help from a mental health professional should symptoms arise.

64:

Secondary risk prevention, including aggressive risk factor management through medical therapy and lifestyle modification, is the mainstay of therapy for patients recovering from acute coronary syndrome. Lifestyle and risk factor modification includes stress reduction, lipid management, optimizing diet and exercise, weight management, and smoking cessation if indicated. Cardiac rehabilitation programs are recommended to improve long-term outcomes, including functional capacity, morbidity and mortality rates, symptom control, and quality of life. Cardiac rehabilitation involves assessment, surveillance, counseling, and training to ensure lifestyle and risk factor modifications are maintained (Arnett et al., J Am Coll Cardiol 2019, 74: 1376-414).

65:

A high-intensity statin should be administered once the patient is stabilized, provided no contraindications are present. Statins reduce the risk of death due to coronary heart disease, recurrent myocardial infarction, revascularization needs, and stroke. Contraindications to statin therapy include hypersensitivity and active liver disease (Benjo et al., Catheter Cardiovasc Interv 2015, 85: 53-60; Amsterdam et al., Circulation 2014, 130: 2354-94; O'Gara et al., Circulation 2013, 127: 529-55).

66:

Follow-up telephone calls, conducted two to four days post discharge, can be an effective means of bridging the inpatient-outpatient transition and provide an opportunity to answer questions as well as address new symptoms and medication related issues (Jack et al., Ann Intern Med 2009; 150(3): 178-187; Kripalani et al., JAMA 2007; 297(8): 831-841; Inglis et al., Cochrane Database Syst Rev 2010, 8: CD007228).

67:

The availability of disease management programs vary. Programs are traditionally focused on common chronic conditions such as ischemic heart disease, diabetes, chronic obstructive pulmonary disease, asthma, and heart failure. Patients who receive education based on disease-specific self-management principles have fewer hospital readmissions, emergency department and unscheduled family physician visits, as well as an increased quality of life (Bourbeau et al., Arch Intern Med 2003; 163(5): 585-591). Transitional care or chronic care management programs can increase compliance with medications and follow-up care, yield higher provider and patient satisfaction, and reduce admissions and emergency department visits (Jackson, Health Affairs 2013, 32: 1407-15).

68:

The medical home provides comprehensive primary care that facilitates partnerships between individual patients, their providers, and when appropriate, the patient's family. The provision of a medical home allows better access to health care, increase satisfaction with care, and improves health (Ziring et al., Pediatrics 1999, 104: 978-81. Reaffirmed 2005). Key attributes of the model include increasing access to healthcare, leveraging long term relationships, and facilitating comprehensive coordinated care. Enrollment in a medical home program has been shown to decrease the risk of readmission (Bayer, Innovations in reducing preventable hospital admissions, readmissions, and emergency room use. 2010; Hines et al., Nurs Econ 2010, 28: 74-85).

69:

A referral should be made to a comprehensive smoking cessation program that includes behavior changing techniques, patient education, and both pharmacological and non-pharmacological interventions (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024).

70:

GOLD guidelines recommend outpatient follow-up with a primary care physician or pulmonologist within 30 days and again at 90 days from hospital discharge in order to prevent hospital readmission and improve health (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; Shah et al., Chest 2016, 150: 916-26).

71:

Starting maintenance medications prescribed after a chronic obstructive pulmonary disease (COPD) exacerbation within 30 days of discharge has been found to decrease future COPD exacerbations and the risk of emergency room visits and hospital admissions (Coutinho et al., Int J Chron Obstruct Pulmon Dis 2016, 11: 1223-31).

72:

Proper inhaler technique is vital to optimizing the delivery of medication to the lungs and minimizing systemic side effects. Incorrect inhaler usage negatively impacts disease control and quality of life (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024).

73:

Prior to discharge following a chronic obstructive pulmonary disease (COPD) exacerbation, influenza and pneumococcal vaccination status should be reviewed. If a patient has not received vaccination prior to discharge, arrangements should be made for outpatient administration (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; Criner et al., Chest 2015, 147: 883-93).

74:

Poor adherence to prescribed medication results in an increase in chronic obstructive pulmonary disease (COPD) symptoms, hospitalization, mortality, cost of medication, and irregular follow-up appointments (Rogliani et al., Respiratory Medicine 2017, 129: 117-23).

75:

Pulmonary rehabilitation after an acute exacerbation of chronic obstructive pulmonary disease (COPD) for 6-8 weeks is associated with improved health-related quality of life, decreased mortality rate, and a lower rate of hospital readmissions (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; Criner et al., Chest 2015, 147: 883-93).

76:

Patients with pneumonia who are being discharged should review influenza and pneumococcal vaccinations. If the patient has not received either vaccination prior to discharge, arrangements should be made for outpatient administration (Mandell et al., Clin Infect Dis 2007; 44 Suppl 2: S27-72; Fiore et al., MMWR Recomm Rep 2010, 59: 1-62; Centers for Disease Control and Prevention. MMWR Recomm Rep 2010, 59: 1102-6; Tomczyk et al., MMWR Morb Mortal Wkly Rep 2014, 63: 822-5).

77:

A referral should be made to a comprehensive smoking cessation program that includes behavior changing techniques, patient education, and both pharmacological and non-pharmacological interventions (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024).

78:

Smoking cessation should be a goal of smokers who are hospitalized with pneumonia, as smoking is an independent risk factor for the development of pneumonia (Aliberti et al., Lancet 2021, 398: 906-19; Mandell et al., Clin Infect Dis 2007; 44 Suppl 2: S27-72).

79:

Reference: (Furie et al., Stroke 2011, 42: 227-76)

80:

Based on the results from the Stroke Prevention by Aggressive Reduction in Cholesterol Levels (SPARCL) trial, high-intensity statin therapy, when appropriate, is recommended for all patients with ischemic stroke and transient ischemic attack to reduce the risk of stroke and cardiovascular events (Grundy et al., Circulation 2019, 139: e1082-

e143).

81:

Social Determinants of Health (SDOH)

Social Determinants of Health (SDOH) are conditions in a person's environment that may affect their health. SDOH factors may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. Patients should be screened for SDOH-related barriers to health to mitigate any unmet social and psychosocial factors impacting the health of the patient. The following SDOH factors should be evaluated on admission to determine which factors and interventions are relevant to an individual patient. This list may not be all inclusive and all relevant SDOH factors should be considered. Any social need determined to be relevant to a given patient should be planned for throughout the hospital stay.

82:

Domestic Abuse (Feltner et al., Jama 2018, 320: 1688-701).

83:

Domestic Abuse and Community-Led Nursing Interventions(Boyle and Murphy-Tighe, J Adv Nurs 2022, 78: 1601-17).

84:

Hospice Care (Noh et al., Am J Hosp Palliat Care 2022, 39: 822-30).

85:

Income Insecurity (Pottie et al., CMAJ 2020, 192: E240-E54).

86:

Food insecurity (Testa and Jackson, Ann Epidemiol 2021, 58: 22-8; Berman et al., Pediatr Rev 2018, 39: 235-46).

87:

A capacity evaluation may be appropriate for a patient who is unable to make decisions for self, refuses treatment or demands to be discharged without a rational reason.

88:

Readmission Risk (Hutchison et al., Prof Case Manag 2022, 27: 47-57).

89:

Risk for Malnutrition (Arensberg et al., Inquiry 2022, 59: 469580221081431).

90:

Risk For Treatment Noncompliance and Mental Illness (National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.).

91:

Tobacco Use (U.S. Department of Health and Human Services, Tobacco Use in Children and Adolescents: Primary Care Interventions. 2021; American Academy of Family Physicians, Tobacco: Preventing and Treating Nicotine Dependence and Tobacco Use (Position Paper). 2019).

92:

Lead Poisoning (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019).

93:

Housing Vulnerability (Anderst et al., BMJ Open 2022, 12: e054338; Martin et al., J Am Board Fam Med 2019, 32: 521-

30; Pottie et al., CMAJ 2020, 192: E240-E54).

94:

Homeless and Vulnerably Housed People (Pottie et al., CMAJ 2020, 192: E240-E54)

95:

Case Management Interventions For The Homeless (Ponka et al., PLoS One 2020, 15: e0230896).

96:

Social Determinants of Health (SDOH) are conditions in a person's environment that may affect their health. SDOH factors may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. Children and caregivers should be screened for SDOH-related barriers to health to mitigate any unmet social and psychosocial factors impacting the health of the child and family unit (Kreuter et al., Annu Rev Public Health 2021, 42: 329-44; Sokol et al., Pediatrics 2019, 144:). The following SDOH factors should be evaluated on admission to determine which factors and interventions are relevant to the individual child and their family support system. This list may not be all inclusive and all relevant SDOH factors should be considered. Any social need determined to be relevant to a given child and family should be planned for throughout the hospital stay. Interventions should be considered for all patients and families if social need is relevant.

97:

Parent Child Programs (Hildebrandt et al., Infant Ment Health J 2020, 41: 677-96; Vanderplasschen et al., Front Psychiatry 2019, 10: 186).

98:

Treatment Considerations for Youth (Substance Abuse and Mental Health Administration, Treatment Considerations for Youth and Young Adults with Serious Emotional Disturbances and Serious Mental Illnesses and Co-occurring Substance Use. 2021).

99:

Parent Training Programs (Nurse-Family Partnership National Resources for Families. 2021; Gubbels et al., Int J Environ Res Public Health 2019, 16:).

100:

Anticipatory Guidance (Dosman and Andrews, Paediatr Child Health 2012, 17: 75-80).

101:

Technology Dependent Children (Pediatrics 2008, 122: 1119-26).

102:

Post-discharge programs include (Nurse-Family Partnership National Resources for Families. 2021; Health Resources & Services Administration, Maternal and Child, The Maternal, Infant, and Early Childhood Home Visiting Program 2021):

- Early Head Start program for infants or toddlers under the age of 3 and pregnant individuals in need of medical, mental health, nutrition and education
- Nurse-Family Partnership for individuals pregnant with their first child and for ongoing nursing visits through the child's second birthday

103:

Behavioral Support (Reyes et al., Health Equity 2021, 5: 100-18; Substance Abuse and Mental Health Administration, Treatment Considerations for Youth and Young Adults with Serious Emotional Disturbances and Serious Mental Illnesses and Co-occurring Substance Use. 2021; Pierron et al., BMC Public Health 2018, 18: 1087).

104:

Lactation Specialist (Leeman et al., *Pediatr Qual Saf* 2019, 4: e130).

105:

Youth Suicide (Milliman et al., *J Pediatr Nurs* 2021, 59: 1-9).

106:

School Based Interventions (Fazel et al., *Lancet Psychiatry* 2014, 1: 377-87).

107:

Children and their caregivers at risk for medication nonadherence may benefit from text message reminders to take medication(s) and the use of electronic medication dispensers.

108:

Peer Support Groups (Mental Health America, *Peer Support: Research and Reports* 2021; National Institute for Health and Care Excellence (NICE). *Psychosis and schizophrenia in adults: prevention and management*. CG178. London: NICE; 2014.).

109:

Telehealth (Solo-Josephson et al., *Telemed J E Health* 2022, 28: 558-65; *J Am Acad Child Adolesc Psychiatry* 2017, 56: 875-93).

110:

Teen Pregnancy (Maness et al., *J Adolesc Health* 2016, 58: 636-43).

111:

Eating Disorders (Gato-Moreno et al., *Int J Environ Res Public Health* 2021, 18;; Al-Khudairy et al., *Cochrane Database Syst Rev* 2017, 6: Cd012691).

112:

Eating Disorders and Family Based Treatment (Lock and La Via, *J Am Acad Child Adolesc Psychiatry* 2015, 54: 412-25).

113:

Childhood Obesity (DeCamp et al., *Health Promot Pract* 2022, 23: 518-29).

114:

Parenting Resources, National Parent Help Line (National Parent and Youth Helpline; U.S. Department of Health and Human Services, Parenting Resources. 2021).

115:

Parenting Resources (U.S. Department of Health and Human Services, Parenting Resources. 2021).

116:

Case Management Interventions for the Homeless (Ponka et al., *PLoS One* 2020, 15: e0230896; Vijverberg et al., *BMC Psychiatry* 2017, 17: 284).

117:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., *Ann Emerg Med* 2013, 62: 145-50; Frei-Jones et al., *Pediatr Blood Cancer*

2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

118:

A visit with a primary care provider within a week of discharge is associated with a 30%(0.30) decrease in Medicaid hospital readmission rates compared to those who had later follow-up visits (Wiest et al., JAMA Netw Open 2019, 2: e187369).

119:

As part of the comprehensive discharge plan, all instruction should be in the patients preferred language, using translators as needed. Instruction materials should be written at an appropriate health literacy level (National Transitions of Care Coalition, 2008-2012). The Agency for Healthcare Research and Quality (AHRQ) provides a toolkit for adult and pediatric practices to help support communication and understanding of health information, navigation of the health care system, and patient self-management. Written materials in some hospitals are written at a third or fourth grade reading level using the patient's native language. They include the reason for hospitalization, symptoms to watch out for, and a plain language list of appointment follow-ups and information related to medications (Boutwell, et al., AHRQ Designing and Delivering Whole-Person Transitional Care: The Hospital Guide to Reducing Medicaid Readmissions. September 2016).

120:

Approximately 40 percent of patients have test results returning after discharge and physicians are commonly unaware of these results. Ten percent of these results require MD intervention (Roy et al., Ann Intern Med 2005; 143 (2): 121-128). Any test with a pending result at discharge holds the potential for an error due to a missed action. Steps must be taken to ensure the appropriate follow-up provider is aware of pending tests and that the results are communicated in a timely manner (Were et al., J Gen Intern Med 2009; 24(9): 1002-1006).

121:

Educational nursing interventions that occur early during hospitalization have been shown to increase the patient's knowledge of their condition as well as self care. Early involvement with self care activities can reduce medication errors and engage the patient in participating in their own care and treatment plan. Teaching should start as soon as the patients and family members are ready to learn (Osborn and Squires, J Ambul Care Manage 2012, 35: 118-28).

122:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

123:

Encouraging patients and caregivers to assume a more active role in improving self-management knowledge such as medication management, condition management, and confidence about plan of care can reduce hospital readmissions (Pediatrics 2008; 122(5): 1119-1126; Coleman et al., Arch Intern Med 2006; 166(17): 1822-1828).

124:

For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., MMWR Recomm Rep 2016, 65: 1-

49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

125:

One in four patients does not adhere to prescribed drug therapy. Poor adherence is considered a critical barrier to treatment success and remains a leading challenge to healthcare professionals (Simpson et al., *Bmj* 2006; 333(7557): 15). Although nonadherence affects all age groups, older adults have specific barriers against effective medication use and can be more vulnerable to the incorrect administration of medication. In persons age 60 years or older, nonadherence with medication regimens varies from 26-59 percent. Reasons for nonadherence include confusion with doses and schedules, forgetfulness in taking the medication, toxic drug interactions, and excessive financial expense leading to underuse (Tangalos, *Annals of Long Term Care* 2006; 14:). Interventions that may successfully improve adherence include patient education, empowerment, reminders, frequent clinic visits, or telephone calls from staff or physicians. Every attempt should be made to simplify the patient's drug regimen by reducing the number of pills per day and by minimizing medication costs wherever possible (Mukherjee, *Am Heart J* 2008; 155 (4): 589-590).

126:

Primary care physician surveys, as well as a checklist tool developed by the Centers for Medicare and Medicaid Services, recommend including the following information in a discharge summary:

- Diagnoses
- Brief reason for hospital stay including the patient's condition on discharge and comorbidities and/or behavioral health conditions, if applicable
- Comorbidities and behavioral health conditions, if applicable
- Medication and allergy lists
- Pending test results
- Copy of patient's advance directive
- Care instructions
- Follow-ups with post-hospital clinical, behavioral, and social services, if indicated
- List of follow-up appointments
- Follow-ups for high risk patients

Transmission of a discharge summary should occur within 48 hours of discharge (Boutwell, et al., *AHRQ Designing and Delivering Whole-Person Transitional Care: The Hospital Guide to Reducing Medicaid Readmissions*. September 2016; Kripalani et al., *JAMA* 2007; 297(8): 831-841).

127:

Ensuring the medical practitioner receives the discharge summary is a key intervention in safely transitioning care and improving outcomes (Jack et al., *Ann Intern Med* 2009; 150(3): 178-187). Summaries should be available at the time the patient is leaving the hospital. If the summary is not available at that time, it should be provided within

48h of discharge (Kripalani et al., JAMA 2007; 297(8): 831-841). As recommended by the National Quality Forum's Safe Practice on Discharge (2006), hospitals should confirm that the patient's medical practitioner received the discharge summary. This requires a tracking mechanism and may be in any format viable in each organization's structure, from a phone call (documented in the medical record) to a secure e-mail or a return fax (Society of Hospital Medicine, Project BOOST. 2008 [cited Nov 2015]).

128:

For those patients covered under a Medicare benefit, the Affordable Care Act mandates a certifying physician or an allowed non-physician practitioner, document that a face-to-face encounter has occurred with the patient prior to certifying eligibility for home care. To ensure this requirement is met for this patient population, the following should be included in the discharge plan:

- Determine who will perform the fact-to-face encounter
- Implement a policy for documenting encounters and transmitting this documentation to home care agencies

(Centers for Medicare and Medicaid Services, 2010)

129:

The Centers for Medicare and Medicaid Services (CMS) recommends discharge planning teams be knowledgeable about local post acute providers, including special services. CMS requires hospitals to analyze readmission data and reassess discharge planning policies on an ongoing basis. Upon admission from a post acute setting, the discharge planning team must determine whether the originating provider is capable of continuing to care for the patient after hospital discharge (Centers for Medicare and Medicaid Services, Hospital Appendix A- Interpretive guidelines for 42 482.43, Discharge planning. 2013).